

Projects in Data Science Aardvark

Group Aardvark

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[Group Aardvark Project](#)

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Contents

1	Introduction	3
1.1	Related work	3
2	Dataset analysis and annotation	4
2.1	Data exploration	4
2.2	Annotations	6
2.3	Preprocessing	6
3	Feature extraction	7
3.1	Asymmetry feature	7
3.2	Border feature	7
3.3	Color feature	8
3.4	Diameter feature	9
4	Baseline classification method	9
4.1	Feature analysis	9
4.2	Classification models	12
4.3	Model evaluation	13
4.4	Best baseline model	14
5	Extended classification method	15
5.1	Hair removal	15
5.2	Hair coverage feature	16
5.3	Annotation analysis	17
5.4	Feature comparison	18
5.5	Best extended model	19
6	Evaluation	21
7	Open question	22
7.1	Idea	22
7.2	Experimental setup	23
7.3	Results	23
7.4	Conclusion	25
8	Discussion	26
9	Final conclusion	26
A	Appendix	29

1 Introduction

The clinical utility of automated skin lesion classification depends heavily on visual analysis of morphological features, diagnostic criteria, and the expertise of the reviewing specialist [5]. Skin cancer is among the most commonly developing malignancies, and its incidence continues to rise, yet early-stage lesions are often difficult to distinguish from benign ones by the naked eye alone [3]. Diagnosis still relies largely on the subjective judgement of dermatologists, whose availability is unevenly distributed and particularly restricted in rural and under-resourced regions [5]. These constraints have motivated strong interest in computer-aided diagnosis systems that can support specialists and make screening more consistent, fast and accessible [4]. Because early detection is closely tied to favourable outcomes, even a modest, reliable, and reproducible automated screening tool carries real clinical value. This project focuses on the binary problem of distinguishing malignant from benign skin lesions in dermoscopic images. Rather than learning directly from raw pixels, we represent each lesion through a small set of interpretable, self-made features computed with image-processing techniques. A central difficulty in this setting is that dermoscopic images are acquired under varied conditions and frequently contain elements such as body hair and pen markings, which are unrelated to the lesion itself but can distort feature extraction and mislead a classifier. A further difficulty is that some clinically intuitive descriptors translate poorly into image measurements: lesion diameter, for instance, is sensitive to the distance and angle at which an image is cap-

tured, an issue that has led other feature-based work to set this descriptor aside [3]. The objective of this work is therefore not only to build a classifier, but to examine how feature design, element handling, and image quality affect its behaviour. Our approach computes features inspired by the ABCD rule - asymmetry, border irregularity, color variation, and diameter - using objective image-processing measures rather than strict clinical definitions, in the spirit of earlier efforts to evaluate these criteria automatically [1]. These features are paired with classical machine learning models to establish a transparent baseline. We then extend this baseline with an artifact-removal preprocessing step and an additional feature describing hair coverage, and we evaluate whether such preprocessing improves classification. The contribution of this work is threefold: a reproducible feature-based pipeline for malignant–benign lesion classification, a systematic comparison of classical classifiers under this representation, and a controlled analysis of how artifact removal and image downscaling affect performance — providing evidence on which preprocessing choices are worthwhile and which add complexity without measurable benefit.

1.1 Related work

The automatic classification of skin lesions has been extensively studied in both medical imaging and machine learning. Traditional approaches often rely on handcrafted features inspired by the ABCDE rule (Asymmetry, Border, Color, Diameter, Evolution), which provides a simple clinical framework for identifying suspicious skin lesions [6]. These criteria are widely used as a first-line

screening tool for early melanoma detection.

In this project, a similar feature-based approach is adopted, focusing on asymmetry, border irregularity, color variation, and diameter. While these features are inspired by the ABCDE framework, they are computed using image processing techniques rather than following strict clinical definitions.

Several studies have pursued this objective, feature-based direction. One study proposed automatic measures for asymmetry, color variegation, and diameter, computing each criterion directly from segmented dermoscopic images rather than relying on subjective clinical assessment [1]. More recently, another approach combined ABCD-inspired preprocessing with learned classifiers, but excluded the diameter criterion on the grounds that inconsistent acquisition distance and angle make it unreliable — a limitation our own analysis also encounters [3]. These works confirm that handcrafted criteria can be quantified objectively, while highlighting that their reliability depends strongly on segmentation quality and image acquisition conditions.

More recent work has shifted towards deep learning approaches, particularly convolutional neural networks (CNNs), which automatically learn hierarchical features directly from image data. These models have demonstrated strong performance on large-scale datasets and, in some cases, have achieved performance comparable to dermatologists [2]. However, such approaches typically require significantly larger datasets and computational resources.

In contrast to purely data-driven deep learning approaches, this project focuses on interpretable handcrafted features combined with classical machine learning models. This

enables a transparent analysis of model behaviour while providing a clear and reproducible baseline for further methodological development.

2 Dataset analysis and annotation

In accordance with the course guidelines, the PAD-UFES-20 dataset was utilized to train our model. This dataset comprises 2298 dermoscopic images collected from 1373 patients, representing a total of 2298 distinct skin lesions. Each image is annotated with one of six clinically defined lesion categories:

- Actinic Keratosis (ACK)
- Basal Cell Carcinoma (BCC)
- Malignant Melanoma (MEL)
- Melanocytic Nevus (NEV)
- Squamous Cell Carcinoma (SCC)
- Seborrheic Keratosis (SEK)

These lesion types can further be grouped into two broader diagnostic classes:

- **Malignant:** BCC, SCC, and MEL
- **Benign:** ACK, NEV, and SEK

2.1 Data exploration

To gain insight into the dataset, we manually inspected three different subsets, referred to as Groups G, I, and O. The objective of this exploration was to better understand

class distributions, image variability, and potential confounding visual factors that may influence downstream model performance.

During inspection, a substantial portion of images were observed to contain visible body hair, while others were entirely hair-free. Additionally, several images included pen markings drawn on the skin surface, whereas others did not. A small subset of images appeared slightly blurry, which may reduce feature clarity and potentially impact the performance of machine learning models.

Variations in lighting conditions were also present across the dataset. While most images were adequately illuminated, some exhibited minor inconsistencies in brightness and contrast, which could influence both visual interpretation and automated analysis. Furthermore, images containing multiple lesions were identified. This may pose challenges for lesion segmentation and subsequent feature extraction, particularly when a single diagnosis is associated with the image.

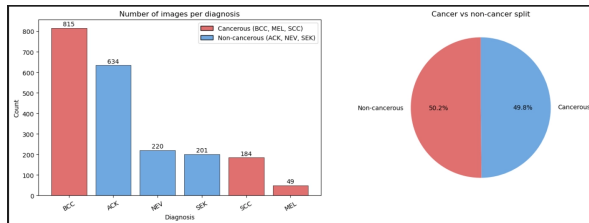


Figure 1: Distribution of images per diagnosis.

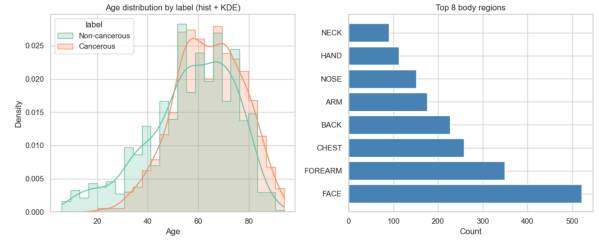


Figure 2: Age distribution by label (left) and top 8 body regions in the dataset (right).

Cancerous lesions tend to appear in older patients and are most frequently located on the face and forearm.

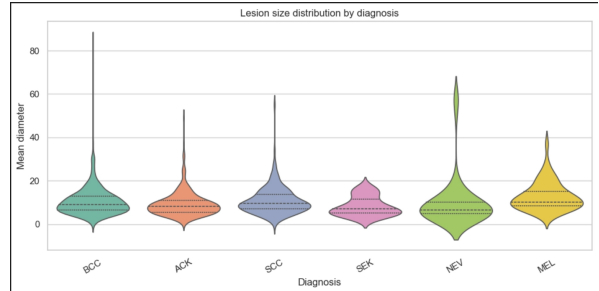


Figure 3: Lesion size distribution by diagnosis.

The median lesion size is similar across all six diagnostic categories, suggesting that diameter alone is not a strong discriminator between malignant and benign lesions. BCC and NEV shows the widest size distribution with the longest upper tail, while SEK remains the most compact.

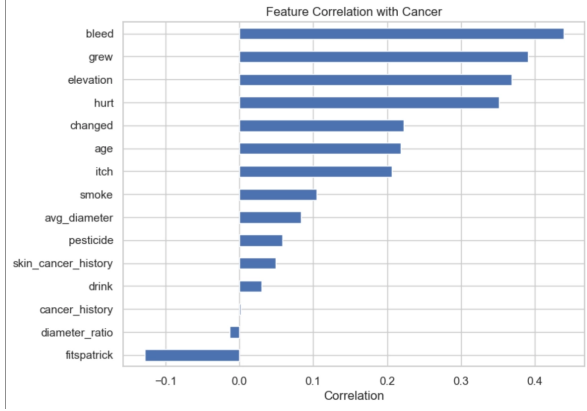


Figure 4: Correlation of patient metadata features with cancer label.

Bleeding shows the strongest positive correlation with malignancy, suggesting it is the most informative clinical symptom for cancer detection. Fitzpatrick skin type has the biggest negative correlation, indicating that lighter skin tones are more associated with malignancy. This is consistent with known epidemiological patterns, as people with lighter skin have less melanin protection and are at higher risk of skin cancers.

2.2 Annotations

To address the presence of visual artifacts in the dataset, manual annotations were created for each image. The annotation process was carried out collaboratively by course participants, who were divided into smaller groups, with each group responsible for annotating a subset of the dataset.

The annotations focused on identifying the presence of body hair and pen markings, as these artifacts are unrelated to the underlying lesion and may negatively affect automated analysis. The amount of visible hair was graded on a discrete scale from 0 to

3, where 0 indicates no visible hair and 3 indicates heavy hair coverage. The presence of pen markings was recorded as a binary variable, where 0 denotes absence and 1 denotes presence.

Artifacts such as hair and pen marks commonly appear in dermoscopic images as a result of clinical examination procedures. If left unaddressed, models may inadvertently learn correlations between these artifacts and diagnostic labels, leading to reduced generalization and potentially misleading predictions. Manual annotation enables explicit identification of these artifacts, allowing them to be removed, masked, or analyzed separately during preprocessing.

2.3 Preprocessing

In this project, we aimed to utilize as many images and corresponding masks as possible. However, a discrepancy was observed, as the dataset contained more images than masks. Since masks are required for feature extraction and no manual segmentation was performed, only images with corresponding masks were included in the analysis.

An additional issue arose from masks containing multiple segmented regions. Since the feature extraction functions are designed to operate on a single lesion, multiple regions could negatively impact the reliability of the computed features. To address this, a filtering procedure was applied based on the following criteria:

1. A mask is considered valid if it contains only one region.
2. If a mask contains multiple regions, it is considered valid only if the largest

region is at least three times larger than the second largest region.

For masks meeting these criteria, only the largest region is retained for feature extraction, under the assumption that it corresponds to the primary lesion. This ensures a consistent and reliable input for the subsequent analysis.



Figure 5: Mask with multiple regions

3 Feature extraction

To represent each lesion quantitatively, we extracted four features based on the well-known ABCDE rule used in dermatology. We decided to use the ABCD only. These features capture key visual characteristics associated with malignancy:

- **Asymmetry:** Measures the degree to which the lesion shape deviates from symmetry.
- **Border irregularity:** Quantifies the smoothness or irregularity of the lesion boundary.
- **Color variation:** Captures the diversity of colors present within the lesion.

- **Diameter:** Represents the size of the lesion.

These features were selected as a starting point, as they are frequently referenced in dermatological contexts and provide a relatively simple way of describing lesion characteristics. Although this representation is limited compared to using full image data, it allows for easier implementation and initial experimentation with different classification methods.

Prior to classification, the extracted features were normalized to ensure that all variables contribute equally to the learning process.

3.1 Asymmetry feature

To compute the asymmetry feature, we apply the “folding method” along the horizontal and vertical axes of the lesion mask. Each axis-fold produces a Jaccard (IoU) score between the two halves; the two scores are averaged into a single symmetry score for the current orientation. To make the measure robust to lesion orientation, we repeat this process for 8 evenly spaced rotation angles between 0° and 180° (in 22.5° steps) and retain the highest score across all angles. When the two halves being compared do not have exactly the same shape (due to odd dimensions or rotation), the smaller half is zero-padded so the Jaccard score can be computed.

3.2 Border feature

Compactness describes how close the shape of a lesion is to a perfect circle. A perfect circle has a compactness of 1.0. The more

irregular, jagged, or complex the shape, the larger its perimeter becomes, and the lower its compactness value

$$\text{Compactness} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$$

Malignant lesions normally have uneven, asymmetrical edges, which result in lower compactness. Benign lesions are more even and rounded, which leads to higher compactness.

Area is the area enclosed by the contour. It is computed using OpenCV (using a geometric method based on Green’s theorem), rather than simple pixel counting.

Perimeter is the length of the contour. It is computed using OpenCV, as the sum of Euclidean distances between consecutive contour points, which makes it more accurate than simply counting edge pixels, as it correctly accounts for diagonal segments.

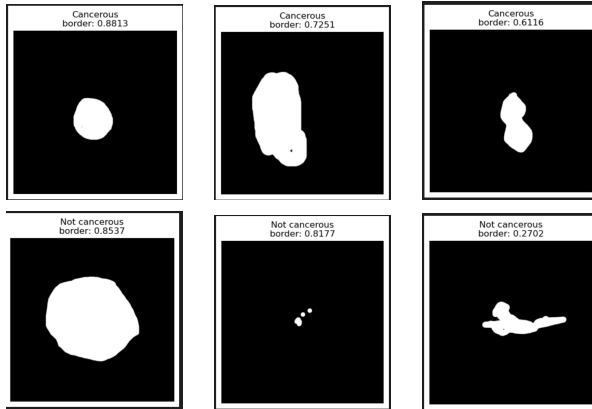


Figure 6: Examples of cancerous (top row) and non-cancerous (bottom row) lesions.

3.3 Color feature

We extracted 12 color-related features using the HSV color space: Hue, Saturation, and

Value. HSV was chosen because it separates color tone from brightness and saturation, which makes it more suitable for describing lesion coloration than raw RGB values.

The first group of features consists of six global HSV statistics computed from all pixels inside the lesion mask. The mean hue, saturation, and value describe the dominant color of the lesion, while the standard deviation of these three channels measures overall color variation. Higher variation can indicate heterogeneous pigmentation, which is often considered relevant when assessing suspicious lesions.

The second group consists of three local color non-uniformity features computed using SLIC superpixels. We first divide the image into small, visually coherent regions and keep only the superpixels that overlap with the lesion mask. For each valid superpixel, we compute its mean HSV values. The standard deviation of these superpixel-level means is then used to describe how much the color changes across different parts of the lesion. This allows us to capture local color irregularity rather than only relying on global pixel statistics.

The third group consists of three relative color contrast features. These features compare the average HSV values inside the lesion with the average HSV values of the surrounding skin. The surrounding skin region is obtained by dilating the lesion mask and subtracting the original mask, resulting in a narrow ring around the lesion. The absolute differences in hue, saturation, and value describe how strongly the lesion stands out from nearby skin.

Since hue is circular in the HSV representation, we use a circular mean when computing average hue values. All feature values are

normalized to comparable ranges. Overall, these features describe three complementary aspects of lesion color: dominant coloration, internal color heterogeneity, and contrast against surrounding skin.

12 chosen features:

- 6 global HSV statistics summarize the lesion’s dominant hue, saturation, and value, plus how much these vary overall
- 3 region-based variation features describe how uneven the lesion’s color is across smaller local regions
- 3 relative difference features describe how different the lesion is from surrounding skin

3.4 Diameter feature

The diameter is computed using a projection-based approach along the horizontal and vertical axes. The number of foreground pixels is counted for each row and column, and the maximum values define the horizontal and vertical extents, respectively. The diameter is then taken as the maximum of these two extents.

To ensure comparability across images with different resolutions, the diameter is normalized by the maximum image dimension. This produces a scale-invariant feature that reflects relative object size rather than absolute pixel measurements. If no valid foreground pixels are present, the diameter is set to NaN.

Since the projections are taken along fixed image axes, this method is not rotation invariant: a lesion rotated 45° relative to the axes has a smaller projected extent than the same lesion aligned with them. In practice

this introduces some variance in the measured diameter - a known limitation of axis-aligned projection methods. We accept this trade-off for the sake of simplicity and robustness; normalization by the image dimension ensures that the feature captures relative lesion size rather than pixel-resolution effects.

4 Baseline classification method

4.1 Feature analysis

Asymmetry feature analysis For the symmetry feature, we used a density plot to gain intuition about how the cancerous population sample differs from the non-cancerous one. We observed substantial overlap between the two groups, so we conducted an appropriate statistical test to validate this intuition. We failed to reject the null hypothesis (no difference between the two groups) at the 0.05 significance level, and therefore could not conclude that the groups differ significantly. This indicates that symmetry has limited discriminative power in this dataset and may not be informative on its own.

Border feature analysis The overall distribution is left-skewed, with a mean of 0.735 and median of 0.776, meaning most lesions are moderately compact but a long left tail of highly irregular shapes exists.

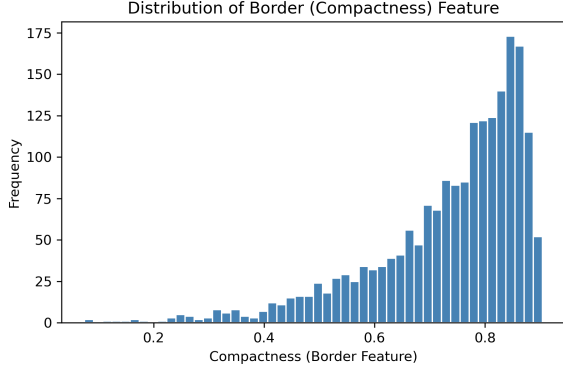


Figure 7: Distribution of Border (Compactness) Feature

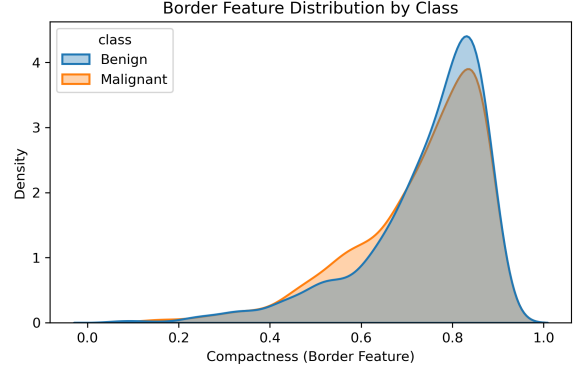


Figure 8: Border Feature Distribution by Class

Benign lesions have a slightly higher mean compactness (0.741) than malignant ones (0.729), consistent with the clinical expectation that cancerous lesions tend to have more irregular, jagged boundaries. A Welch’s t-test confirms that this difference is statistically significant ($t = -2.04$, $p = 0.042$), and this result is corroborated by the non-parametric Mann-Whitney U test ($p = 0.042$). However, the effect size is very small (Cohen’s $d = -0.09$), and the Pearson correlation between border and the binary cancer label is only -0.046 . This means that while the difference is real, the border feature alone has very limited discriminative power.

Color feature analysis The t-test results show that several color-related features differ significantly between benign and malignant lesions. The strongest differences were observed for saturation variation features. In particular, malignant lesions had higher values for both global saturation standard deviation and superpixel-based saturation

standard deviation. This suggests that malignant lesions tend to show greater saturation heterogeneity, which may reflect more irregular pigmentation.

Mean hue, mean saturation, mean value, and value standard deviation were also statistically significant after Bonferroni correction, although their effect sizes were smaller. This indicates that these features may contribute some information to the classifier, but they do not separate the classes strongly on their own. The relative value difference showed a negative mean difference, meaning that benign lesions had a higher brightness contrast with the surrounding skin than malignant lesions in this dataset.

Overall, the t-tests suggest that color features contain useful information, especially saturation-based heterogeneity features. However, the effect sizes and density plots show substantial overlap between benign and malignant lesions. Therefore, color features are helpful but not sufficient for reliable classification on their own. In addition, the correlation matrix showed that global

and superpixel-based saturation variation are highly correlated, suggesting that some color features may be redundant.

Feature	Benign mean	Malignant mean	Mean diff.	Corrected p -value	Cohen's d
hsv_std_s	0.078	0.101	0.023	1.34×10^{-12}	0.643
sp_hsv_std_s	0.070	0.090	0.019	8.70×10^{-17}	0.595
hsv_mean_h	0.631	0.658	0.027	4.05×10^{-14}	0.361
rel_hsv_diff_v	0.067	0.050	-0.017	1.07×10^{-8}	-0.282
hsv_mean_s	0.337	0.372	0.035	1.18×10^{-6}	0.243
hsv_mean_v	0.602	0.626	0.024	1.23×10^{-3}	0.177
hsv_std_v	0.076	0.081	0.005	2.14×10^{-2}	0.142

Table 1: Welch’s t-test results for the color-related features that remained significant after Bonferroni correction. Positive mean differences indicate higher average values for malignant lesions, while negative values indicate higher average values for benign lesions.

Figure 9 shows the density distributions of the 12 color-related features for benign and malignant lesions. Overall, the distributions overlap substantially, indicating that no single color feature provides a clear separation between the two classes. However, some features show visible differences. In particular, the standard deviation of saturation and the superpixel-based standard deviation of saturation are shifted toward higher values for malignant lesions. This suggests that malignant lesions may contain more variation in saturation, which can be interpreted as a form of color heterogeneity.

The mean saturation feature also shows a small shift, with malignant lesions tending to have slightly higher saturation values. In contrast, hue-based features, such as mean hue, hue standard deviation, superpixel hue standard deviation, and relative hue difference, show very similar distributions for both classes. This could mean that hue alone is not highly discriminative in our dataset.

The relative color contrast features show limited class separation. In particular, relative value difference has a longer tail for benign lesions, meaning that some benign lesions also differ strongly from surrounding skin in brightness. Therefore, contrast with surrounding skin is not necessarily specific to malignant lesions. Overall, the density plots indicate that color features may contribute useful information to the classifier, especially saturation-based heterogeneity features, but they are unlikely to be sufficient for reliable classification on their own.

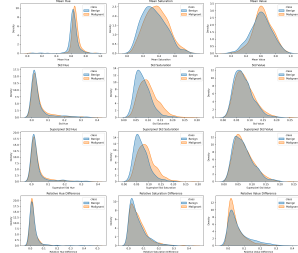


Figure 9: Distribution plots of each color feature

The correlation matrix shows that several color features are strongly correlated, especially the global HSV standard deviation features and their corresponding SLIC-based superpixel features. This indicates that both approaches capture similar aspects of color heterogeneity within the lesion. In particular, hue, saturation, and value variation measured globally are almost identical to the variation measured across superpixels. The relative contrast features are moderately correlated with some variation features, suggesting that lesions with stronger internal color irregularity also tend to differ more from the surrounding skin. Overall, the matrix suggests that the color feature set contains redundant information, and that

feature selection may be useful to reduce overfitting.

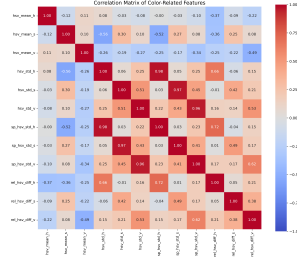


Figure 10: Correlation matrix of color-related features

To conclude the analysis, the correlation matrix suggests that some color features are similar and may contain redundant information. However, after testing a reduced feature set where some of these correlated features were removed, the model performance decreased. Therefore, although the analysis indicates potential redundancy, we decided to keep the full feature set because the removed features still appeared to contribute useful information to the classifier.

Diameter The diameter feature represents the relative size of the lesion after normalization. To assess its ability to distinguish between classes, we compare its distribution for benign and malignant lesions.

Figure 11 shows that both classes exhibit very similar distributions, with median values close to 1 and comparable spread. No clear separation is observed, indicating that the diameter feature has limited discriminative power.

This is supported by a statistical test yielding a p -value of 0.906, confirming that the difference between the two groups is not significant.

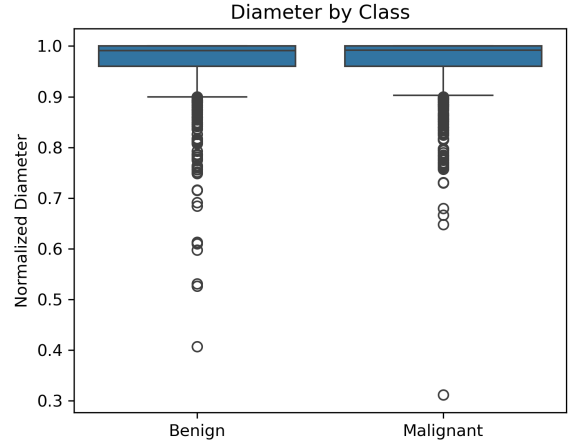


Figure 11: Distribution of the diameter feature for benign and malignant lesions.

4.2 Classification models

To evaluate the effectiveness of the extracted features, three standard classification models were implemented: k-Nearest Neighbors (KNN), Logistic Regression, and Decision Tree classifiers. These models were chosen to provide a range of perspectives, from simple linear models to more flexible non-linear approaches.

Logistic Regression serves as a baseline linear model, assessing whether the features are linearly separable. The Decision Tree classifier enables non-linear decision boundaries and provides interpretability through rule-based decisions. The KNN classifier, which assigns labels based on proximity in feature space, is particularly suitable for low-dimensional representations such as the ABCD features.

To ensure a realistic evaluation and avoid data leakage, the dataset was split at the patient level. All images belonging to a single patient were assigned exclusively to either

the training or test set. An approximate 80/20 split was applied. Model performance was evaluated on the held-out test set.

4.3 Model evaluation

The three classifiers were evaluated on the held-out test set (404 lesions from previously unseen patients) using six complementary criteria: overall test-set metrics, ROC curves, confusion matrices, calibration plots, bootstrap 95% confidence intervals on the AUC, and paired statistical comparisons across the cross-validation folds.

Test-set metrics. Table 2 reports the per-model performance on the held-out set.

Model	Accuracy	Precision	Recall	F1	AUC
kNN	0.676	0.712	0.689	0.700	0.728
Logistic Regression	0.663	0.719	0.635	0.675	0.755
Decision Tree	0.678	0.698	0.730	0.714	0.696

Table 2: Test-set classification metrics for the baseline pipeline.

Logistic Regression achieves the highest AUC and precision, while the Decision Tree reaches the highest recall.

ROC curves. Figure 12 shows the ROC of all three classifiers. Logistic Regression dominates the upper-left region.

Confusion matrices. Figure 13 reports the per-class errors. Logistic Regression produces the fewest false positives (55) but the most false negatives (81); the Decision Tree shows the opposite pattern (70 FP, 60 FN); kNN sits in between.

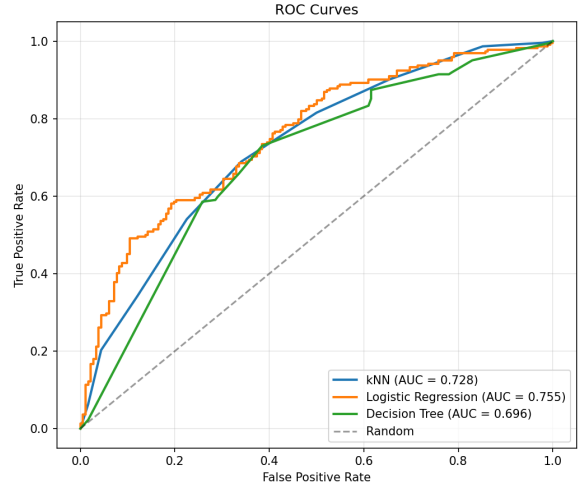


Figure 12: ROC curves for the three baseline classifiers.

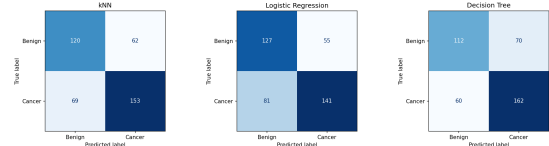


Figure 13: Confusion matrices for kNN, Logistic Regression, and Decision Tree on the baseline test set.

Calibration. Figure 14 plots the empirical fraction of positives against the mean predicted probability. Logistic Regression follows the diagonal most closely in the mid-to-high range relevant for clinical decisions. The Decision Tree shows a downward kink at the highest bin - a typical artefact of pure-leaf splits.

Bootstrap confidence intervals. The test set was resampled with replacement 1 000 times and the AUC recomputed (Table 3, Figure 15). Logistic Regression has

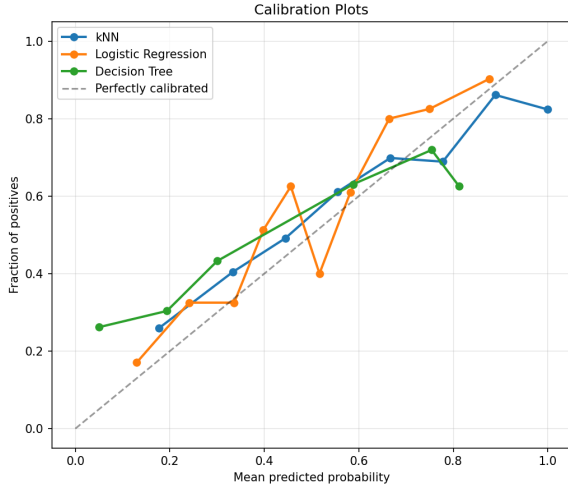


Figure 14: Calibration plots for the three baseline classifiers.

the narrowest interval, indicating the most stable estimate.

Model	AUC (mean)	95% CI	Width
kNN	0.728	[0.678, 0.778]	0.100
Logistic Regression	0.754	[0.706, 0.797]	0.091
Decision Tree	0.696	[0.646, 0.744]	0.099

Table 3: Bootstrap 95% confidence intervals for test-set AUC.

Cross-validation: per-fold AUC. Table 4 reports the per-fold AUC for the best hyperparameter of each classifier.

Pairwise significance tests. Because the three classifiers share the same five CV folds, comparisons are paired. We report both the paired Wilcoxon signed-rank test (primary, non-parametric) and the paired t -test (supplementary), with Bonferroni correction over the three pairwise comparisons (adjusted $\alpha = 0.05/3 = 0.0167$).

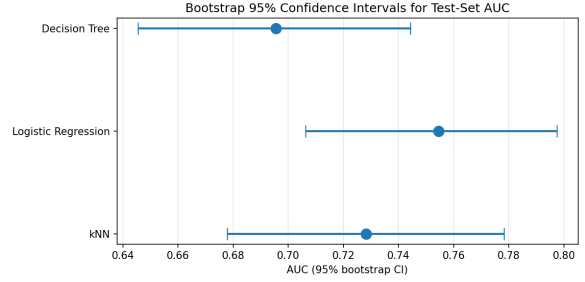


Figure 15: Bootstrap 95% confidence intervals for the AUC of the three baseline classifiers.

Model	F1	F2	F3	F4	F5	Mean	Std
kNN	0.736	0.713	0.709	0.703	0.681	0.709	0.020
LR	0.799	0.794	0.736	0.729	0.765	0.765	0.032
DT	0.766	0.721	0.706	0.673	0.722	0.718	0.034

Table 4: Per-fold cross-validation AUC. F1-F5 = fold 1-5.

At $n = 5$ CV folds, the Wilcoxon test has a mathematical lower bound of $p = 0.0625$; both LR vs. kNN and LR vs. DT reach this floor, meaning LR was higher than the other model on *every* fold. The paired t -test confirms two robust effects: Logistic Regression significantly outperforms kNN ($p = 0.011$) and the Decision Tree ($p = 0.004$) even after Bonferroni correction. No significant difference was detected between kNN and the Decision Tree.

4.4 Best baseline model

Combining all the evidence from Section 4.3, Logistic Regression is selected as the best baseline classifier:

- Highest discriminative performance: test AUC = 0.755; CV mean AUC = 0.765.

Comparison	Δ AUC	Wilc. p	t -test p	Significant?
kNN vs LR	-0.0562	0.0625	0.0110	t , Bonf [†]
kNN vs DT	-0.0090	0.4375	0.5104	-
LR vs DT	+0.0471	0.0625	0.0040	t , Bonf [†]

Table 5: Paired comparison of CV AUC.

[†]Significant under Bonferroni-corrected $\alpha = 0.0167$ (t -test). The Wilcoxon test cannot reach $p < 0.05$ at $n = 5$ folds (floor = 0.0625).

- Most stable estimate: narrowest bootstrap CI (width 0.091).
- Statistically significant over both alternatives: after Bonferroni correction, LR beats kNN ($p = 0.011$) and the Decision Tree ($p = 0.004$); the Wilcoxon test reaches its $n = 5$ floor for both comparisons, indicating LR was the highest model on every fold.
- Best calibration in the clinically relevant range.
- Interpretability: fitted linear coefficients reveal which features push the prediction toward each class.

LR has the lowest recall (0.635), missing 81 of 222 cancer cases on the test set, compared with 60 misses for the Decision Tree. This is the standard precision-recall trade-off at the default 0.5 decision threshold; in a screening setting where missed cancers are more costly than false alarms, the threshold can be lowered to recover recall. Crucially, the threshold-independent AUC - which measures the model’s ability to rank cancer cases above benign ones - is highest for LR.

5 Extended classification method

While the baseline model relies solely on features extracted from the original images, real-world dermoscopic images often contain visual artifacts such as hair and pen markings. These artifacts can interfere with feature extraction, particularly for color- and texture-based features, and may negatively impact classification performance.

To address this, an extended classification approach was developed that incorporates a preprocessing step to remove such artifacts prior to feature extraction.

5.1 Hair removal

Hair detection is performed using morphological image processing techniques. The image is first converted to grayscale, after which both black-hat and top-hat transformations are applied. These operations highlight thin, elongated structures corresponding to dark and light hair, respectively. The resulting masks are combined to produce a binary hair mask representing likely hair locations.

Once detected, hair is removed using an inpainting technique. Inpainting replaces the detected hair pixels by interpolating information from the surrounding area, effectively reconstructing the underlying lesion appearance. This results in a cleaner image where the lesion is less obstructed.

By removing hair through inpainting, these non-lesion pixels are replaced with values interpolated from the surrounding tissue. This results in a smoother and more uniform color distribution that better reflects

the intrinsic properties of the lesion. Consequently, the extracted color descriptors become more consistent and less influenced by external artifacts.

Overall, hair removal is expected to improve the discriminative power of color features by reducing noise and ensuring that the measured color variation corresponds to actual biological characteristics rather than occlusions.

In addition to hair, blue pen markings are also detected and removed. These are identified by thresholding the image in the HSV color space to isolate typical blue color ranges. Similar to hair removal, the detected pen regions are removed using inpainting, thereby restoring the original lesion appearance as much as possible.

5.2 Hair coverage feature

In addition to removing hair, a quantitative measure of hair presence is computed. Specifically, the proportion of lesion area covered by hair is calculated by comparing the number of hair pixels within the lesion mask to the total lesion area.

The resulting value represents the relative extent of occlusion and serves as an additional feature. Higher coverage indicates that a lesion is more affected by artifacts, which may reduce the reliability of extracted features.

Figure 16 illustrates the detection pipeline, where hair is identified and intersected with the lesion mask to compute coverage.

Figure 17 shows that the hair coverage feature increases with annotated hair severity, confirming that it captures the degree of occlusion in a consistent way. While some

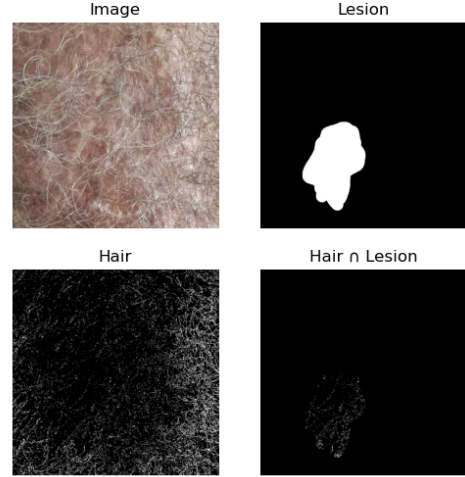


Figure 16: Illustration of hair detection and coverage estimation.

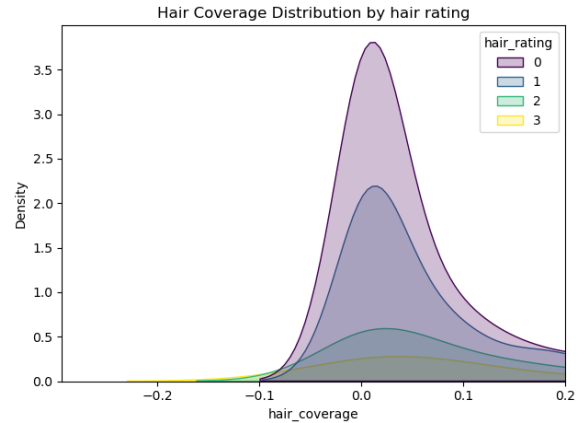


Figure 17: Distribution of hair coverage across different annotation levels. Higher ratings correspond to increased coverage.

overlap between groups is observed, reflecting variability in annotations, the overall trend demonstrates that the feature provides a meaningful and continuous representation of hair presence.

The extended model follows the same pipeline as the baseline, with the addition of preprocessing and the optional hair coverage feature. The same core features are extracted, ensuring that any performance differences can be attributed to the preprocessing step.

5.3 Annotation analysis

To evaluate the reliability and consistency of the annotations, inter-annotator agreement was assessed for both hair and pen mark annotations provided by five annotators. Multiple complementary metrics were used:

- Naïve agreement (pairwise agreement)
- Cohen’s Kappa (pairwise agreement adjusted for chance)
- Fleiss’ Kappa (overall agreement across all annotators)

Pairwise agreement between annotators was computed using both naïve agreement and Cohen’s Kappa. Naïve agreement measures the proportion of samples for which two annotators assign the same label. Cohen’s Kappa extends this by accounting for agreement occurring by chance, providing a more robust estimate of consistency. In Figure 18 we can see a very clear difference between the two different approaches of agreement. In general, the kappa score

are lower than the naïve agreement calculation; kappa penalizes the matching score by chance.

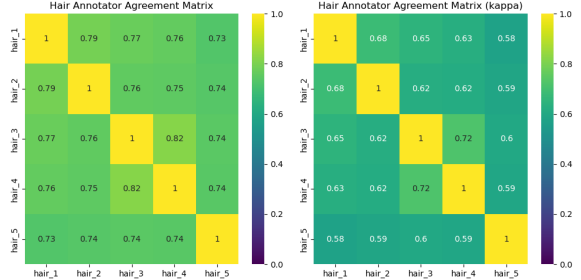


Figure 18: Hair annotator agreement heatmap.

A similar trend is observed in pen mark annotations, where Cohen’s Kappa scores (Figure 19) are consistently lower than naïve agreement, reflecting the adjustment for chance agreement. Overall, pen annotations exhibit higher agreement compared to hair annotations. This is expected, as pen marks are binary (presence or absence), making them easier to identify consistently, whereas hair annotations are ordinal and depend on subjective assessment of hair coverage across the image, leading to greater variability among annotators.

Beyond pairwise agreement, we also computed Fleiss’ Kappa, which measures overall agreement across all five annotators simultaneously. The hair annotations achieved $\kappa_{\text{Fleiss}} = 0.627$, indicating substantial agreement, while the pen annotations reached $\kappa_{\text{Fleiss}} = 0.816$, indicating almost perfect agreement, according to the standard interpretation scale where 0.81-1.00 indicates almost perfect agreement and 0.61-0.80 substantial agreement. This confirms the same trend observed in the pairwise comparisons:

pen marks are easier to identify consistently than hair, whose ordinal scale and subjective nature lead to more variation across annotators.

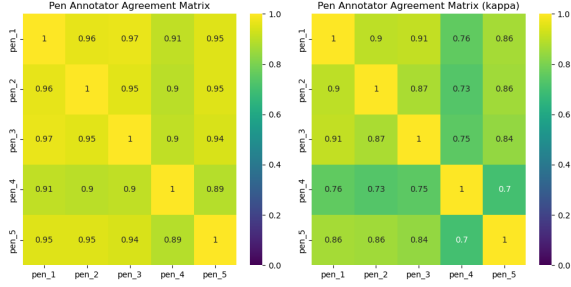


Figure 19: Pen annotator agreement heatmap.

Furthermore, we analyzed variability in hair ratings across the five annotators by examining the relationship between hair coverage and annotator standard deviation. Higher standard deviation indicates greater disagreement, while lower values reflect stronger consensus. As shown in Figure 20, a threshold (red dashed line) is used to distinguish high-variance (blue) and low-variance (red) cases. The results indicate that disagreement is more pronounced for images with low hair coverage, as evidenced by a clear cluster of high-variance points in the low-coverage region.

5.4 Feature comparison

To evaluate the effect of hair removal on feature extraction, we compare feature distributions between the baseline and extended pipelines.

Figure 21 shows that hair removal primarily affects color-related features. In particular, HSV-based variance features (e.g.,

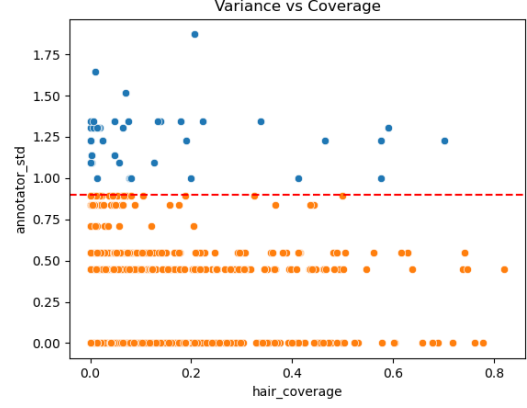


Figure 20: Variance vs Coverage.

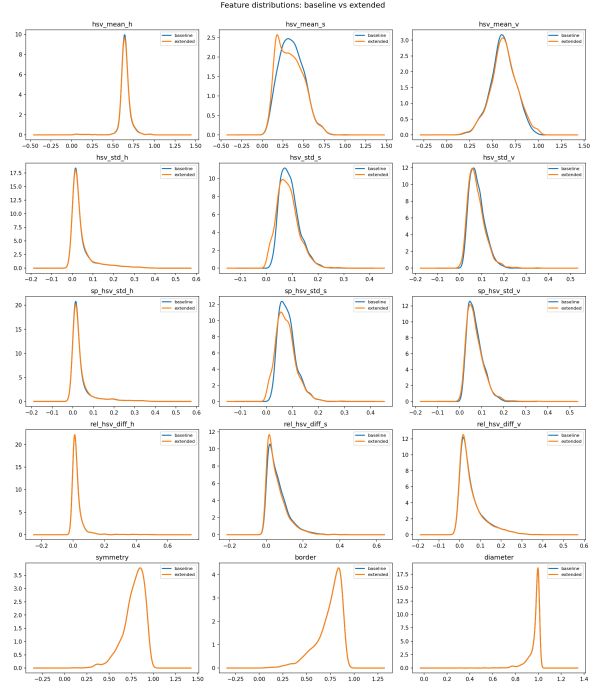


Figure 21: Comparison of feature distributions for baseline (blue) and extended (orange) models. Color-related features exhibit reduced variability after preprocessing, while shape-based features remain largely unchanged.

`hsv_std_*` and `sp_hsv_std_*`) become more concentrated, indicating reduced variability. This suggests that hair introduces artificial intensity variations, which are removed through preprocessing. In contrast, shape-based features such as symmetry, border irregularity, and diameter remain unchanged, as they are computed from the lesion mask and are not affected by image content.

Feature	Mean change	Max change	% changed
<code>hsv_mean_s</code>	0.019	0.517	40.05
<code>hsv_mean_v</code>	0.013	0.365	40.46
<code>hsv_std_s</code>	0.009	0.206	40.15
<code>sp_hsv_std_s</code>	0.007	0.218	40.61
<code>rel_hsv_diff_s</code>	0.007	0.309	40.26
<code>symmetry</code>	0.000	0.000	0.00
<code>border</code>	0.000	0.000	0.00
<code>diameter</code>	0.000	0.000	0.00

Table 6: Summary of feature changes between baseline and extended models. Color-related features are affected in approximately 40% of samples, while shape-based features remain unchanged.

Table 6 confirms that color features are the most affected by preprocessing, with consistent changes across a substantial portion of the dataset. In contrast, the absence of change in shape features serves as a sanity check, verifying that the preprocessing step does not alter the underlying lesion geometry.

To assess statistical significance, a Wilcoxon signed-rank test was performed on the paired baseline and extended features, with Bonferroni correction applied for multiple comparisons ($\alpha = 0.0033$).

The statistical results confirm that most color-related features are significantly affected by hair removal, even after correcting for multiple comparisons. This demonstrates that the observed changes are not

Feature	Mean change	p-value	Significant (Bonf.)
<code>hsv_mean_s</code>	0.019	< 0.001	Yes
<code>hsv_mean_v</code>	0.013	< 0.001	Yes
<code>hsv_std_s</code>	0.009	< 0.001	Yes
<code>sp_hsv_std_s</code>	0.007	< 0.001	Yes
<code>rel_hsv_diff_v</code>	0.004	0.838	No
<code>symmetry</code>	0.000	-	No
<code>border</code>	0.000	-	No
<code>diameter</code>	0.000	-	No

Table 7: Wilcoxon signed-rank test results comparing baseline and extended features. Most color features show statistically significant changes, while shape features remain unaffected.

due to random variation but reflect a consistent effect of preprocessing. Meanwhile, the lack of change in shape features further supports that the preprocessing step selectively improves pixel-level information without distorting structural properties.

Overall, these findings indicate that hair removal reduces noise in color features while preserving geometric characteristics, leading to more reliable feature representations for classification.

5.5 Best extended model

Because Logistic Regression was the strongest classifier on the baseline pipeline (Section 4.4), we restrict the extended evaluation to the same model: the only variable that changes between the two configurations is the feature-extraction pipeline (with vs. without hair removal). All training, validation, hyperparameter tuning (C selection by 5-fold patient-level CV) and the patient-level test split are identical to the baseline setup.

Test-set metrics. Table 8 compares the held-out test-set performance of LR trained on the two feature sets.

Metric	baseline LR	extended LR	Δ (ext – base)
Accuracy	0.6634	0.6485	−0.0149
Precision	0.7194	0.7041	−0.0153
Recall	0.6351	0.6216	−0.0135
F1	0.6746	0.6603	−0.0144
AUC	0.7548	0.7326	−0.0222

Table 8: Test-set metrics for Logistic Regression on the two feature pipelines. All metrics drop slightly under the extended pipeline.

Hair removal reduces every metric, with the largest drop in AUC (−0.022). No metric improves.

ROC curves. Figure 22 overlays the ROC of both versions. The baseline curve sits at or above the extended curve across almost the entire range, with the gap most visible in the high-sensitivity / low-FPR region most relevant for screening.

Confusion matrices. The error structure is almost identical (Figure 23): extended LR makes 3 more false positives (58 vs. 55) and 3 more false negatives (84 vs. 81) than baseline LR, on the same test set of 404 lesions. Hair removal therefore makes the model marginally worse *everywhere*, not better in any particular axis.

Calibration. Both models remain reasonably well calibrated (Figure 24). The two curves track each other closely across the probability range, with no systematic improvement under hair removal.

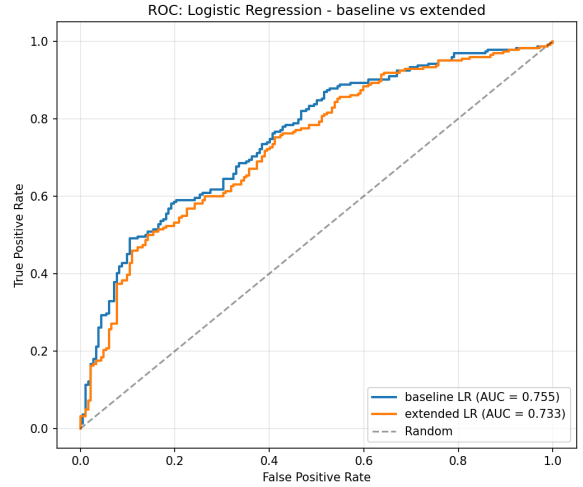


Figure 22: ROC curves: Logistic Regression trained on baseline vs. extended features.

Bootstrap confidence intervals. Re-sampling the test set 1 000 times produces the intervals in Table 9 and Figure 25. The two CIs overlap heavily (baseline: [0.706, 0.797]; extended: [0.683, 0.777]), confirming that the gap in point estimates is well within sampling noise.

Model	AUC (mean)	95% CI	Width
baseline LR	0.7545	[0.7063, 0.7975]	0.0912
extended LR	0.7322	[0.6828, 0.7774]	0.0946

Table 9: Bootstrap 95% confidence intervals for the test-set AUC of LR on each pipeline.

Per-fold paired comparison. Because both pipelines use the same five CV folds, fold i contains the same patients in both, so the comparison is properly paired (Table 10, Figure 26). Extended LR is lower in four of five folds; the single fold where extended was higher (fold 4) shows a comparable gap

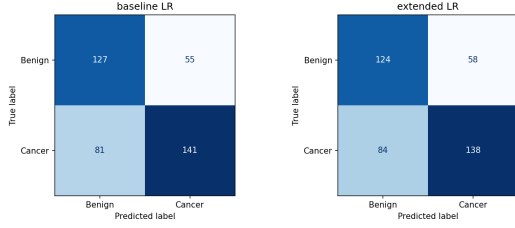


Figure 23: Confusion matrices for Logistic Regression on the baseline (left) and extended (right) pipelines.

in the opposite direction.

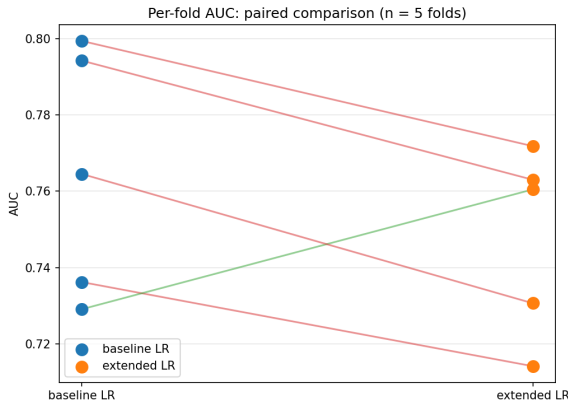


Figure 26: Per-fold AUC, paired by CV fold. Red lines: extended worse; green: extended better.

Model	F1	F2	F3	F4	F5	Mean
baseline LR	0.7994	0.7943	0.7362	0.7291	0.7645	0.7647
extended LR	0.7718	0.7630	0.7141	0.7605	0.7306	0.7480
Δ	-0.028	-0.031	-0.022	+0.031	-0.034	-0.0167

Table 10: Per-fold CV AUC for LR on each pipeline, with paired differences.

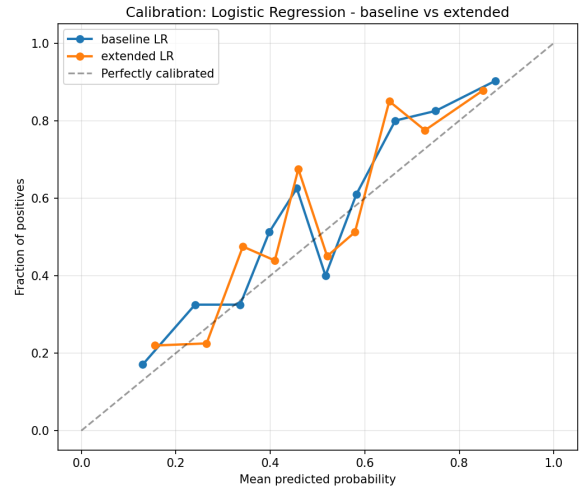


Figure 24: Calibration plots for Logistic Regression on the two pipelines.

Statistical test. There is only one comparison here, so no Bonferroni correction is required. The paired Wilcoxon signed-rank test gives $p = 0.4375$ and the paired t -test $p = 0.2430$ - the difference is not statistically significant. As before, at $n = 5$ folds the Wilcoxon p cannot drop below 0.0625 even in principle. The non-significant t -test result confirms the same conclusion: the modest mean drop in extended (-0.017 AUC) is statistically indistinguishable from noise.

6 Evaluation

The extended pipeline (with hair removal) was hypothesised to improve performance by reducing artefact-driven noise in colour features. The data does not support this:

- Every test-set metric drops slightly under the extended pipeline, including the most important (AUC: 0.755 $\rightarrow$ 0.733).

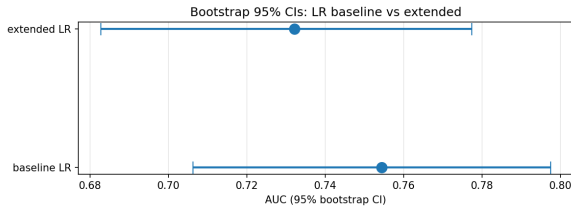


Figure 25: Bootstrap 95% confidence intervals for test-set AUC: baseline vs. extended LR.

- Four of five CV folds get worse, only one improves.
- Bootstrap CIs overlap heavily ([0.706, 0.797] vs. [0.683, 0.777]) and paired tests are not significant ($p = 0.24$, t -test).

The mechanism behind this small drop is documented in `feature_comparison.ipynb` and summarised here: an inpainting-based hair-removal step alters the HSV colour features (most strongly the saturation channel) for the ≈ 800 images annotated as containing hair. Because LR relies heavily on these colour features and inpainting replaces real lesion pixels with interpolated values, the modification injects noise without providing useful signal - consistent with the small but consistently negative effect on classification performance.

We therefore adopt the baseline pipeline with Logistic Regression as the final model. The choice is supported by three converging arguments:

1. **Empirical:** LR on baseline features is at least as good as LR on extended features on every metric we measured.

2. **Statistical:** the paired tests provide no evidence in favour of the extended pipeline; the consistent negative direction of the effect is more compatible with a small real disadvantage of hair removal than with chance.

3. **Methodological:** the baseline pipeline is simpler, faster (no hair-removal step), and has no dependency on the per-image hair annotations, which were only available for a subset of images. Under Occam’s razor, the simpler configuration is preferred when no significant gain comes from adding complexity.

The final model is therefore Logistic Regression trained on baseline features, with cross-validated $\text{AUC} = 0.765$ and held-out test $\text{AUC} = 0.755$ (95% bootstrap CI [0.706, 0.797]).

7 Open question

7.1 Idea

Does Image Downscaling Affect Skin Lesion Classification Performance? The images in the dataset vary substantially in resolution, quality, and file size. Some images are relatively large, which increases the computational cost of feature extraction. To address this, we downscaled all images to a common format before extracting features. This preprocessing step was motivated by two main reasons. First, downscaling substantially improved computational efficiency. Without downscaling, the feature extraction pipeline required more than 20 minutes to run, whereas the downscaled version reduced

the runtime to less than 5 minutes. This made repeated experiments, debugging, and model evaluation more practical within the scope of the project. Second, we wanted to investigate whether reducing image resolution affects the performance of the classification models. This is relevant because skin lesion images may be acquired under different conditions, especially in non-clinical settings where users may take photographs with varying camera quality, lighting, focus, or distance to the lesion. If downsampled images produce classification results comparable to those obtained from higher-resolution images, this would suggest that the selected features are relatively robust to reduced image quality. On the contrary, if performance decreases substantially after downscaling, this would indicate that image resolution and acquisition quality are important factors for reliable lesion classification. Therefore, this experiment evaluates the trade-off between computational efficiency and classification performance. In particular, we examine whether the reduction in processing time can be achieved without a meaningful loss in model performance, measured through the same evaluation metrics used for the baseline classifier. The baseline pipeline configuration was used to evaluate downscaling, since it outperformed the extended pipeline (Section 6).

7.2 Experimental setup

To isolate the effect of image resolution, we built a third feature-extraction pipeline (`open_question`) that differs from the baseline in exactly one step: every image is downsampled so that its largest dimension is 256 px before feature extraction. No hair removal or

pen-mark removal is applied - this matches the baseline preprocessing in every other respect so that the comparison is single-variable. Logistic Regression - the best baseline classifier (Section 4.4) - is then trained on the resulting feature CSV with the same hyperparameter-tuning procedure as in Section 4.3. Because both pipelines share the same patient-level fold split (`random_state = 42`), fold i contains the same patients in both versions and the cross-validation comparison is properly paired.

7.3 Results

Test-set metrics. Table 11 compares the held-out test-set performance of LR on full-resolution vs. downsampled features.

Metric	baseline LR	open_question LR	Δ (oq - base)
Accuracy	0.6634	0.6460	-0.0173
Precision	0.7194	0.6965	-0.0229
Recall	0.6351	0.6306	-0.0045
F1	0.6746	0.6619	-0.0127
AUC	0.7548	0.7325	-0.0222

Table 11: Test-set metrics for Logistic Regression on full-resolution (baseline) and downsampled (`open_question`) features. Downscaling reduces every metric, with the largest drop in AUC (-0.022).

ROC curves. The two ROC curves (Figure 27) closely track each other; the baseline curve sits slightly above the downsampled curve through most of the operating range.

Confusion matrices. The error structure is essentially unchanged (Figure 28): downscaling produces 6 more false positives (61 vs. 55) and 1 more false negative (82 vs.

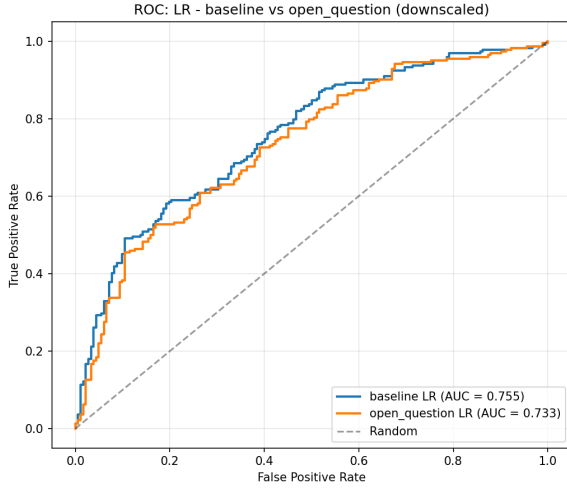


Figure 27: ROC curves: Logistic Regression on full-resolution vs. downsampled features.

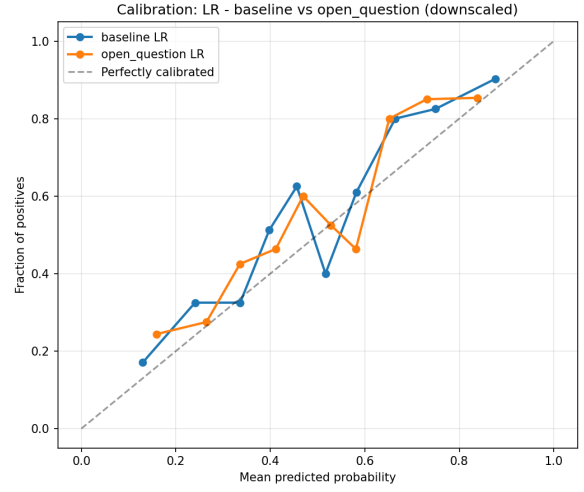


Figure 29: Calibration plots for LR on the two pipelines.

81). Resolution reduction does not shift the precision-recall balance, it shifts both sides of the prediction quality slightly.

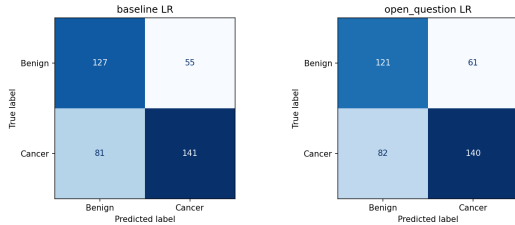


Figure 28: Confusion matrices for LR on the baseline (left) and downsampled (right) features.

Calibration. Both models remain well calibrated (Figure 29); the two curves overlap closely. Downscaling does not damage the meaningfulness of LR’s probability outputs.

Bootstrap confidence intervals. Re-sampling the test set 1000 times yields the intervals in Table 12 and Figure 30. The two CIs overlap heavily (baseline: [0.706, 0.798]; downsampled: [0.683, 0.780]), indicating that the observed gap in point estimates is within sampling noise.

Model	AUC (mean)	95% CI	Width
baseline LR	0.7545	[0.7063, 0.7975]	0.0912
open_question LR	0.7322	[0.6828, 0.7797]	0.0969

Table 12: Bootstrap 95% confidence intervals for LR test-set AUC on each pipeline.

Per-fold paired comparison. The five CV folds are paired (same patient subsets in both pipelines), allowing a direct comparison fold by fold (Table 13, Figure 31). Downscaling reduces AUC in four of five folds; the only fold where downscaling was higher (fold 4) shows a comparable gap in

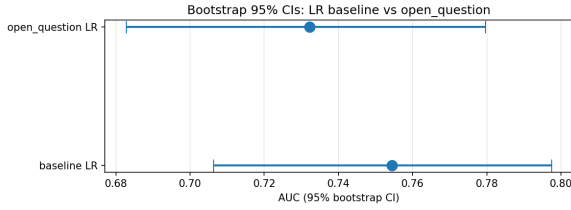


Figure 30: Bootstrap 95% CIs for the test-set AUC of LR on each pipeline.

the opposite direction. The mean fold-level difference is -0.018 AUC.

Model	F1	F2	F3	F4	F5	Mean
baseline LR	0.7994	0.7943	0.7362	0.7291	0.7645	0.7647
open_question LR	0.7549	0.7724	0.7157	0.7655	0.7270	0.7471
Δ	-0.045	-0.022	-0.021	+0.036	-0.038	-0.0176

Table 13: Per-fold CV AUC for LR on each pipeline, with paired differences.

Statistical test. With a single comparison no Bonferroni correction is needed. The paired Wilcoxon signed-rank test gives $p = 0.3125$ and the paired t -test $p = 0.2844$ - the difference is not statistically significant. At $n = 5$ folds the Wilcoxon p cannot in principle drop below 0.0625, and the non-significant t -test result confirms the same conclusion: the modest mean drop of -0.018 AUC is statistically indistinguishable from noise.

7.4 Conclusion

Downscaling images to a maximum dimension of 256 px produces a small but consistently negative effect on classification performance: AUC drops by 0.022 on the held-out test set and by 0.018 on average across CV

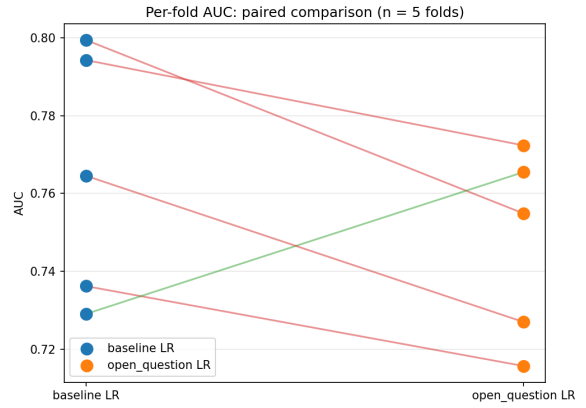


Figure 31: Per-fold AUC, paired by CV fold. Red lines: downscaling worse; green: downscaling better.

folds, with four of five folds favouring the full-resolution pipeline. None of these differences are statistically significant (t -test $p = 0.28$; Wilcoxon $p = 0.31$), and the bootstrap confidence intervals overlap heavily.

The practical implications are nuanced:

- For deployment efficiency: the drop is small enough that downscaling can be a reasonable engineering choice. The original feature extraction takes roughly 20 min on the full dataset, while the downscaled version completes in under 5 min - a four-fold speedup at the cost of 0.02 AUC. For applications where rapid throughput matters more than the last few percentage points of discrimination, downscaling is acceptable.
- For clinical/user-acquired imagery: the consistent direction (four of five folds worse) suggests that high-resolution detail does carry some signal for hand-crafted colour features. Smartphone

photographs - which are essentially downscaled relative to clinical dermoscopes - can be expected to perform slightly worse than dermoscopic images, but the magnitude of the gap is small.

- For our final model: since the open-question experiment shows no advantage for downscaling and a small consistent disadvantage, our final pipeline remains the baseline configuration (no hair removal, no downscaling), with Logistic Regression as the classifier. The downscaled pipeline serves as a useful sensitivity analysis demonstrating that the choice of resolution is not critical.

This open-question result complements the main pipeline-selection result of Section 6: both the addition of hair removal and the reduction of image resolution produce small negative effects, individually not significant, but consistent in direction. Together they argue for the simplest preprocessing pipeline (the baseline) as the most reliable choice for our hand-crafted feature-based classifier.

8 Discussion

Several limitations of the proposed pipeline arise from feature design, preprocessing choices, model selection, and evaluation.

The handcrafted features vary in usefulness: color features are informative, but the diameter contributes less because cropping and normalisation strip away the absolute size information, echoing the difficulty other feature-based studies report when acquisition conditions make diameter unreliable

[3], and suggesting that area-based size measures could be more discriminative.

The pipeline is also sensitive to preprocessing - cropping removes absolute size information, and hair removal, while helpful, can introduce smoothing effects. This suggests that alternative preprocessing strategies, such as less aggressive cropping or improved hair detection methods, could lead to more reliable features, given that artifacts such as hair are well-recognised sources of noise in dermoscopic feature extraction [5].

In addition, the models used are relatively simple and rely entirely on handcrafted features, so they cannot capture more complex patterns that deep models learn directly from image data, often at cost of larger datasets and reduce interpretability [4]. Using more advanced models, such as ensemble methods or learning-based approaches, could improve performance.

Finally, the evaluation setup has limitations. Using only five cross-validation folds, statistical tests have limited power, and small differences are hard to detect. Increasing the number of folds or using repeated cross-validation would provide more reliable results. In addition, using a fixed decision threshold may not be optimal, especially in a medical context where recall is important.

Overall, both feature design and preprocessing strongly shape performance, and targeted improvements in these areas could meaningfully enhance the pipeline.

9 Final conclusion

This project developed and evaluated a feature-based pipeline for the binary classification of skin lesions as malignant or

benign, using the PAD-UFES-20 dataset. Four handcrafted features inspired by the ABCD dermatological rule - asymmetry, border compactness, colour variation, and normalised diameter - were extracted from segmented dermoscopic images and used to train three classical classifiers: k-Nearest Neighbours, Logistic Regression, and a Decision Tree.

Among the four features, colour-based descriptors proved the most discriminative, with saturation heterogeneity showing the strongest and most consistent separation between classes (Cohen’s d up to 0.64). Border compactness showed a statistically significant but practically negligible difference between classes ($d = -0.09$). Asymmetry and diameter showed no significant difference, with the diameter feature particularly affected by the image-cropping and normalisation pipeline.

Logistic Regression was selected as the best baseline classifier, achieving a test AUC of 0.755 and a cross-validated AUC of 0.765, and significantly outperforming both kNN and the Decision Tree after Bonferroni correction. Its advantages include the highest discriminative performance, the best calibration, the narrowest bootstrap confidence interval, and interpretability through its linear coefficients.

Two preprocessing extensions were evaluated against this baseline. First, hair removal via morphological inpainting produced small but consistently negative effects on every metric ($\Delta\text{AUC} = -0.022$), with the drop statistically non-significant (t -test $p = 0.24$) but directionally consistent across four of five folds. Second, downscaling images to 256 px before feature extraction reduced runtime from over 20 minutes to under

5 minutes, at the cost of a similarly small AUC drop ($\Delta\text{AUC} = -0.022$, $p = 0.28$). Neither preprocessing step improved performance, and both were rejected in favour of the simpler baseline pipeline.

The final model is therefore Logistic Regression trained on full-resolution baseline features, with test AUC = 0.755 (95% bootstrap CI [0.706, 0.797]). The results suggest that for handcrafted ABCD-style features, artifact removal and aggressive downscaling add complexity without measurable benefit. Future work could explore area-based size measures to replace the unreliable diameter feature, feature selection to reduce redundancy in the colour descriptor set, and ensemble or learning-based classifiers to capture non-linear patterns beyond the reach of the current models.

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A Appendix

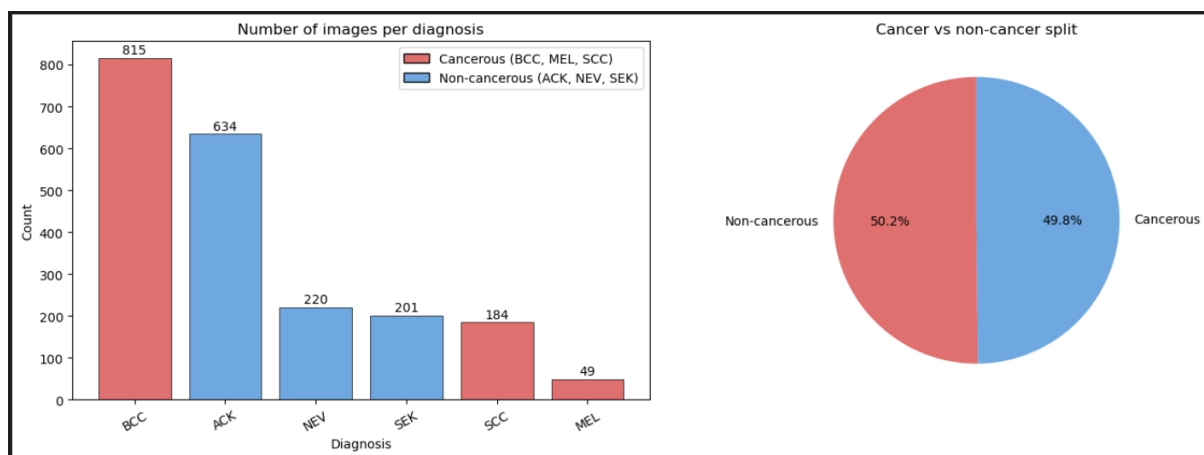


Figure A.1: Distribution of images per diagnosis.

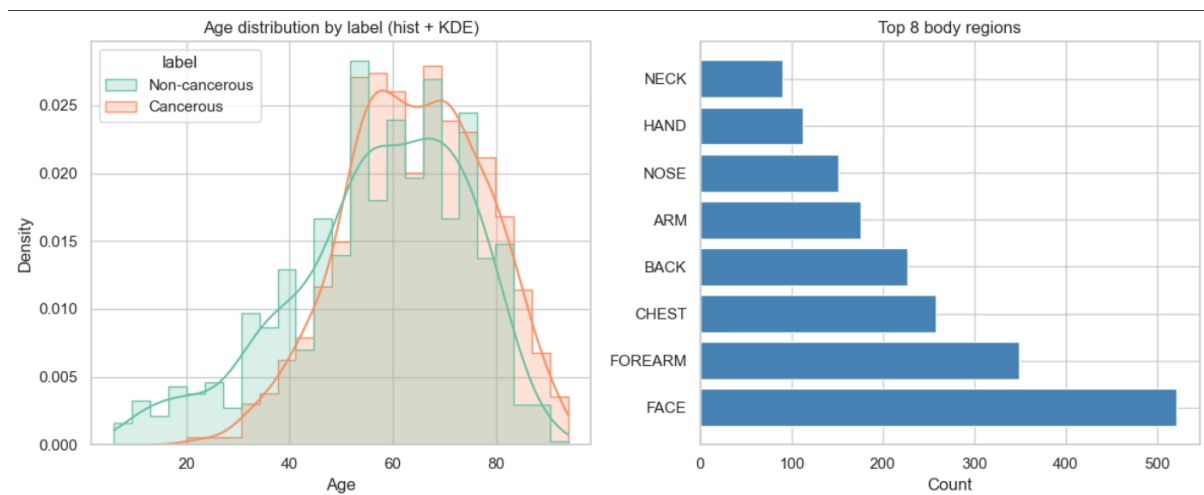


Figure A.2: Age distribution by label (left) and top 8 body regions in the dataset (right).

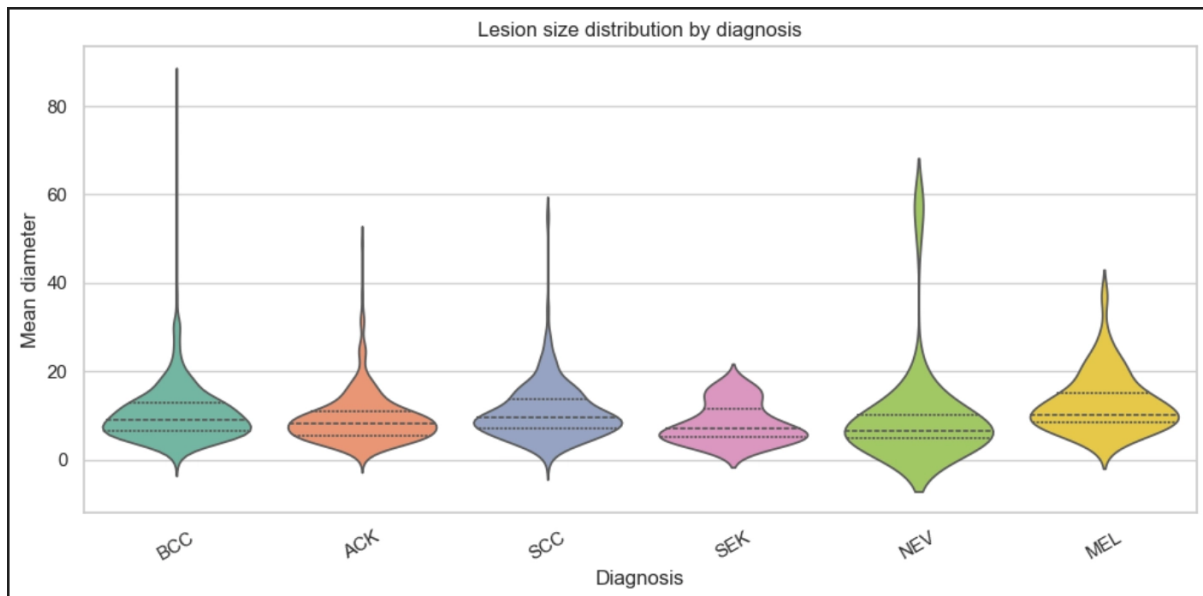


Figure A.3: Lesion size distribution by diagnosis.

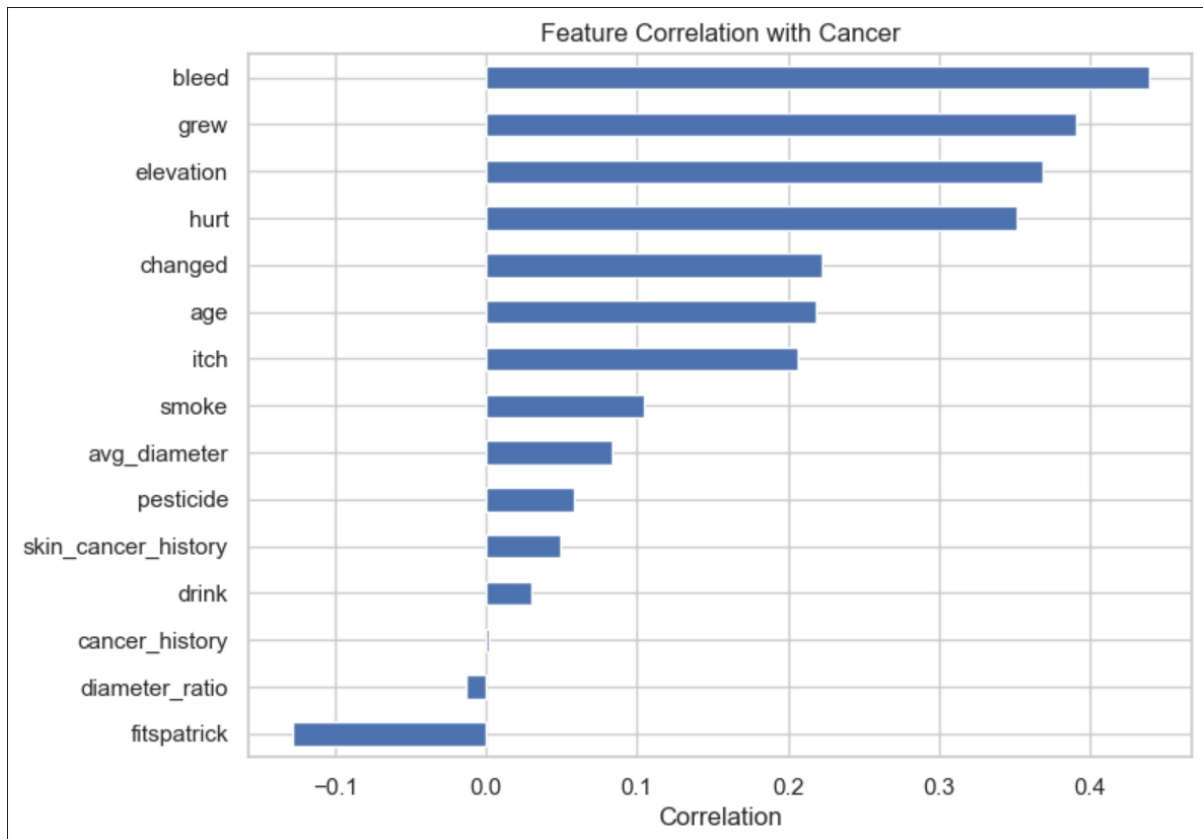


Figure A.4: Correlation of patient metadata features with cancer label.

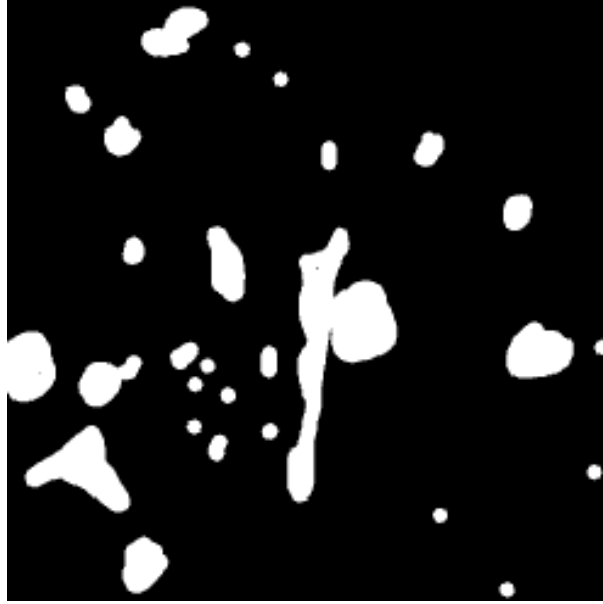


Figure A.5: Mask with multiple regions

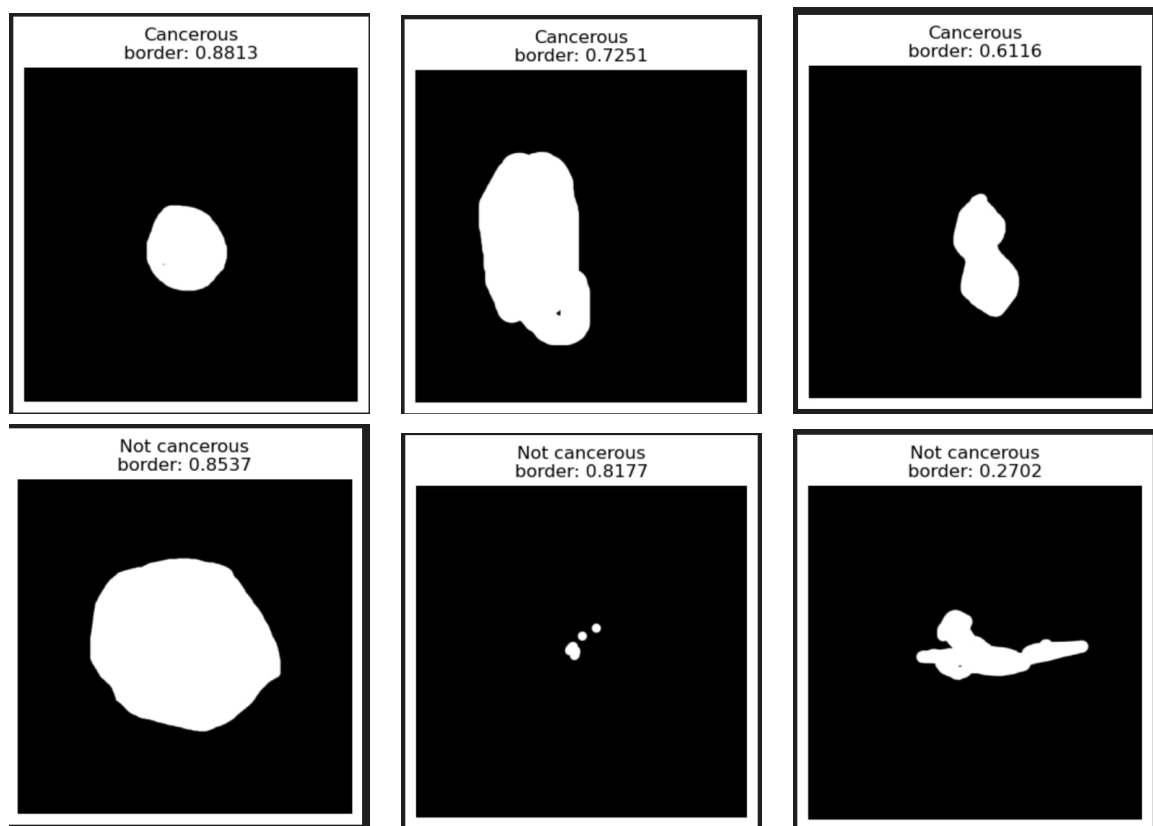


Figure A.6: Examples of cancerous (top row) and non-cancerous (bottom row) lesions.

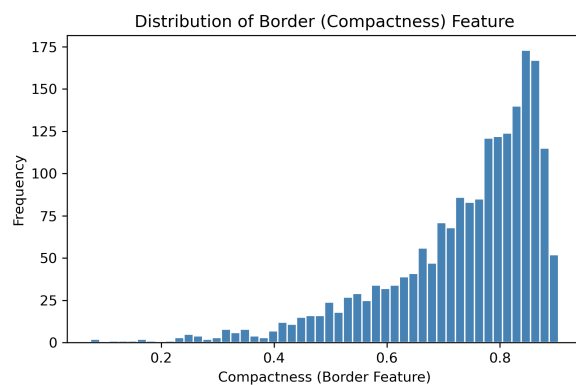


Figure A.7: Distribution of Border (Compactness) Feature.

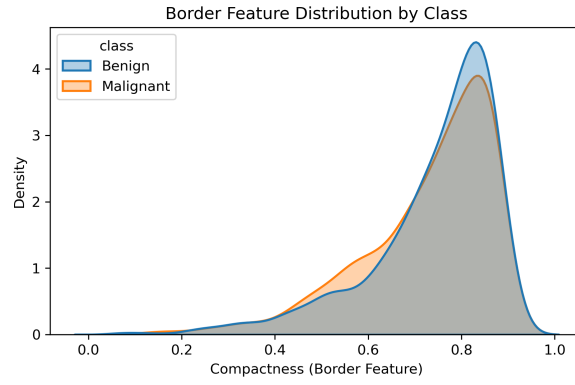


Figure A.8: Border Feature Distribution by Class.

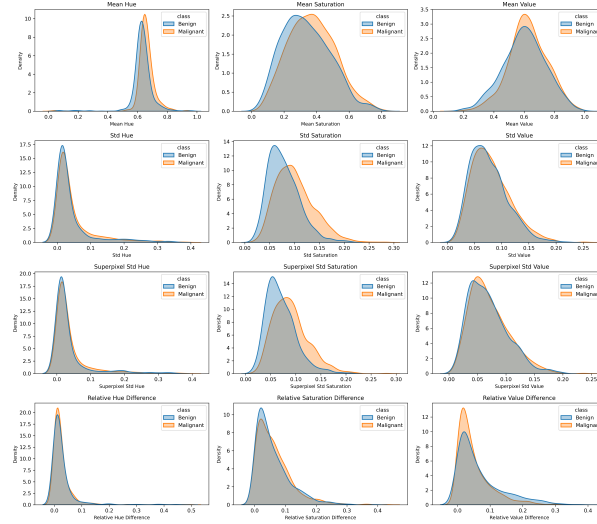


Figure A.9: Distribution plots of each color feature

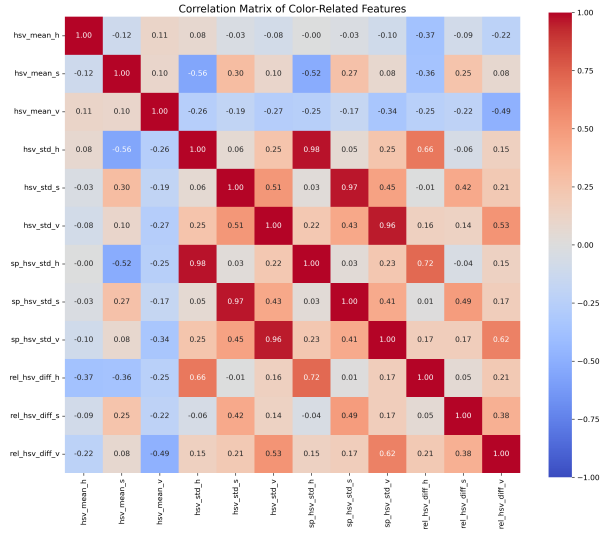


Figure A.10: Correlation matrix of color-related features

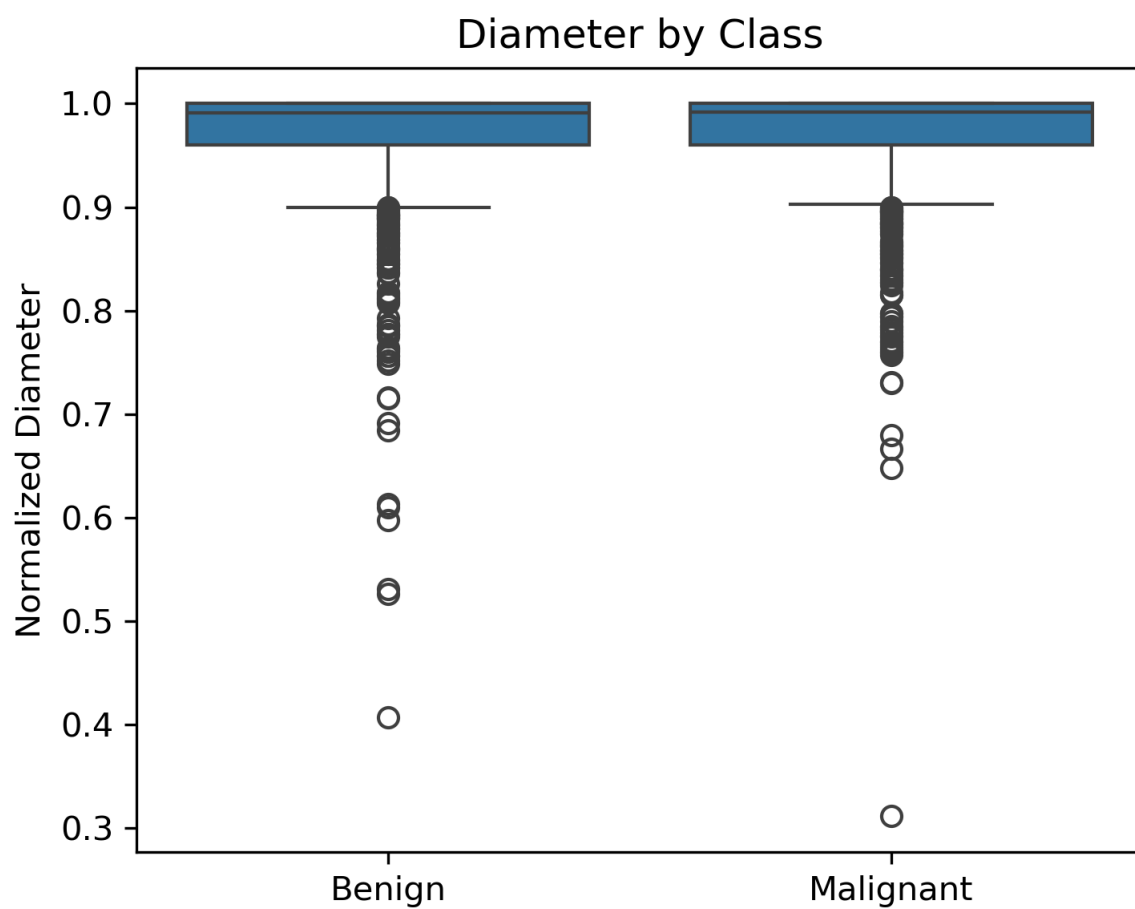


Figure A.11: Distribution of the diameter feature for benign and malignant lesions.

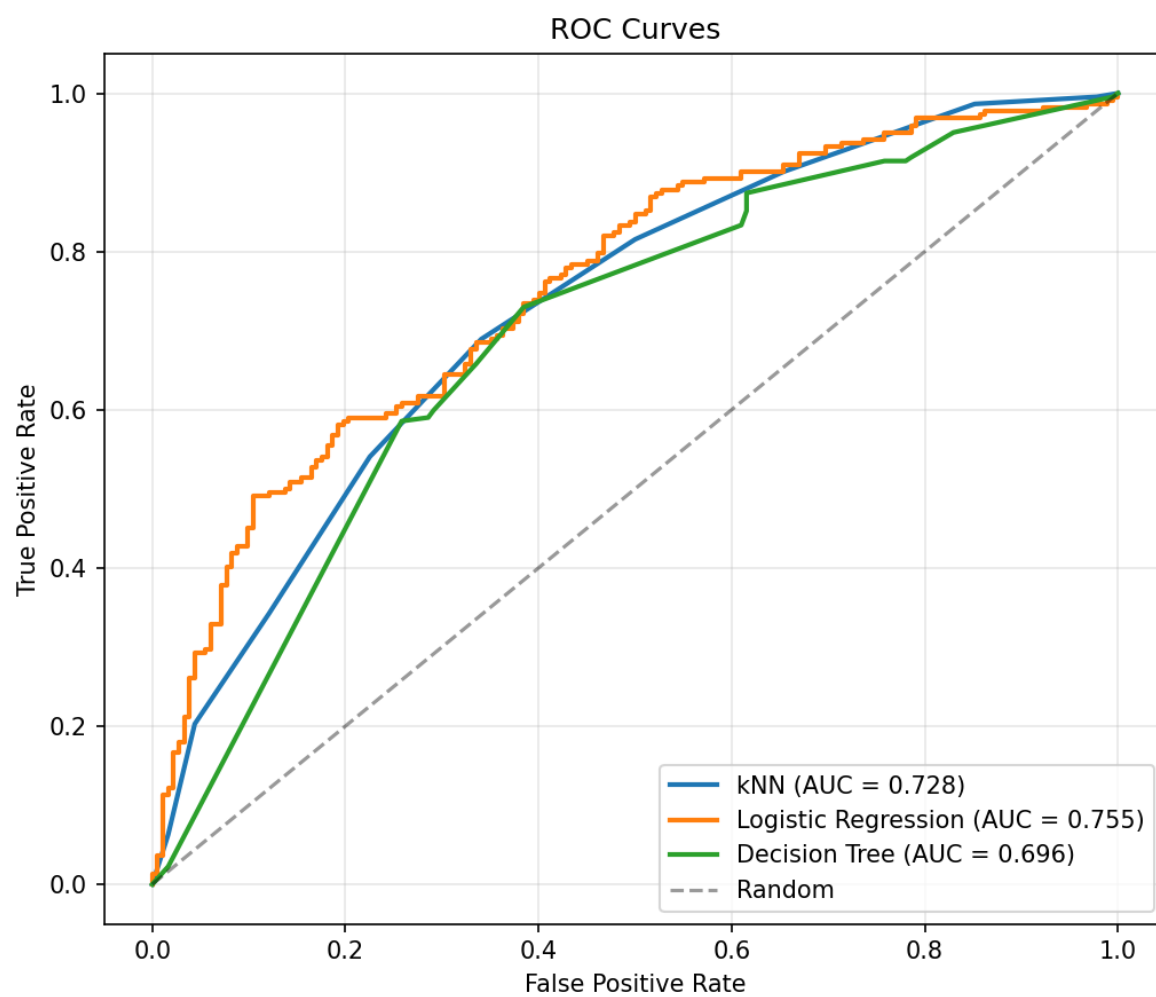


Figure A.12: ROC curves for the three baseline classifiers.

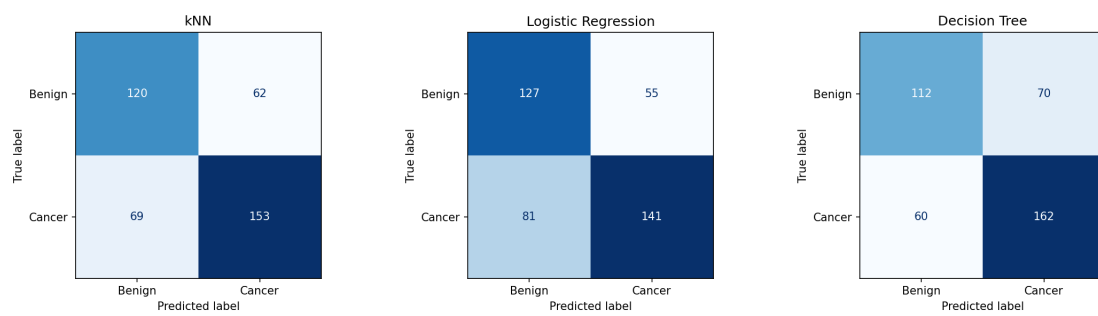


Figure A.13: Confusion matrices for kNN, Logistic Regression, and Decision Tree on the baseline test set.

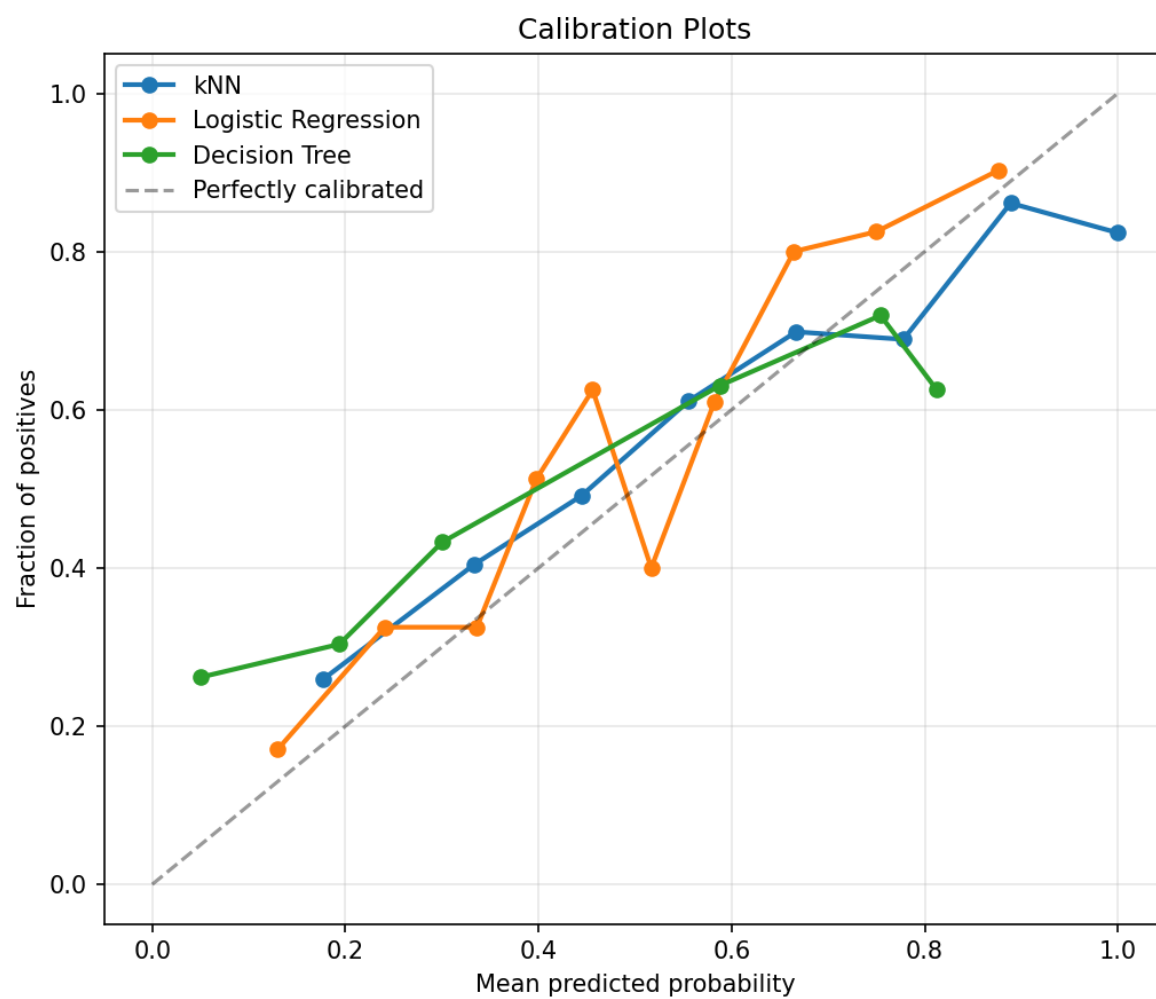


Figure A.14: Calibration plots for the three baseline classifiers.

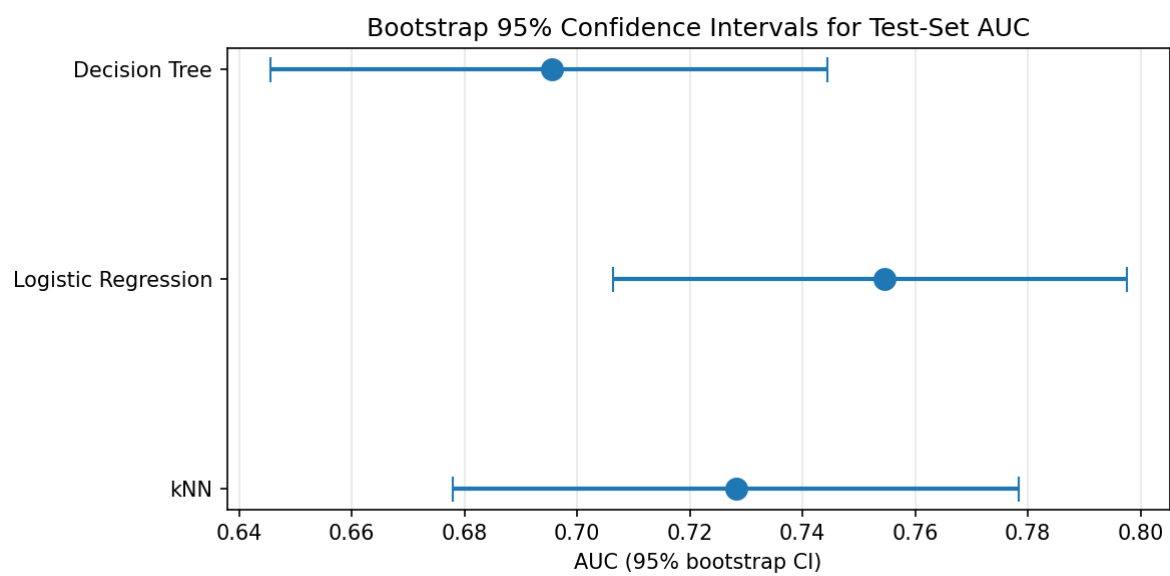


Figure A.15: Bootstrap 95% confidence intervals for the AUC of the three baseline classifiers.

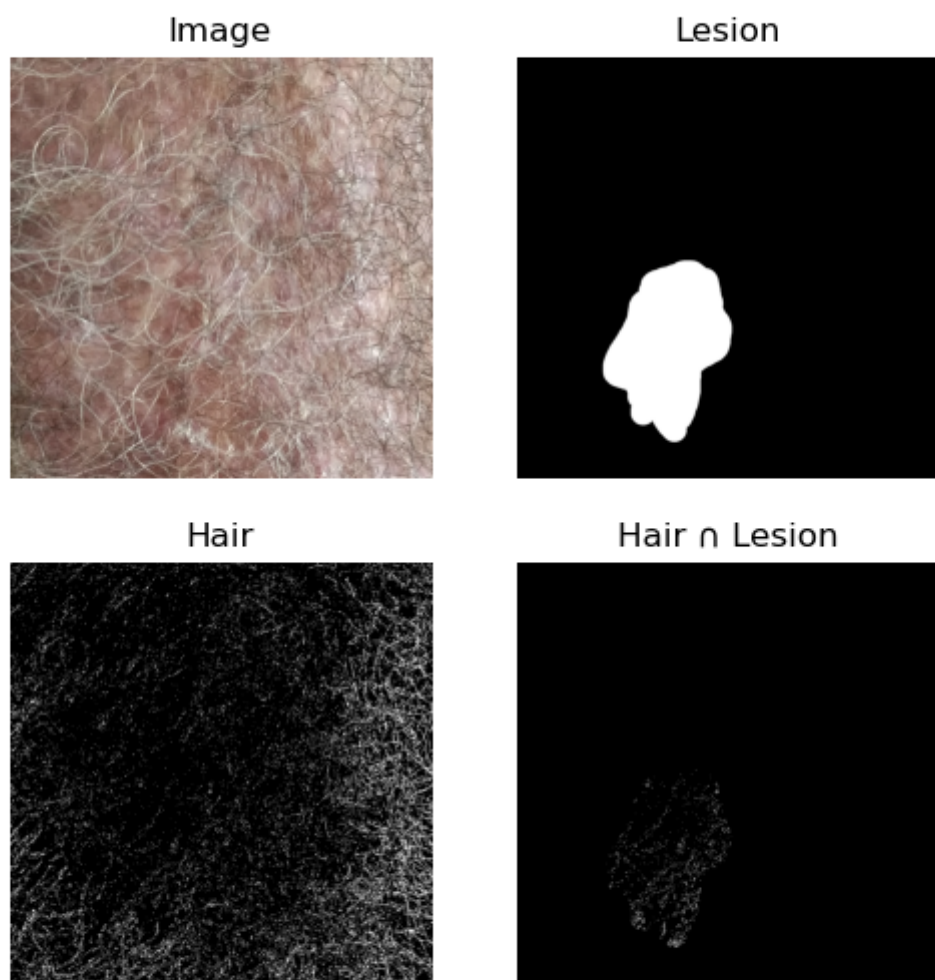


Figure A.16: Illustration of hair detection and coverage estimation.

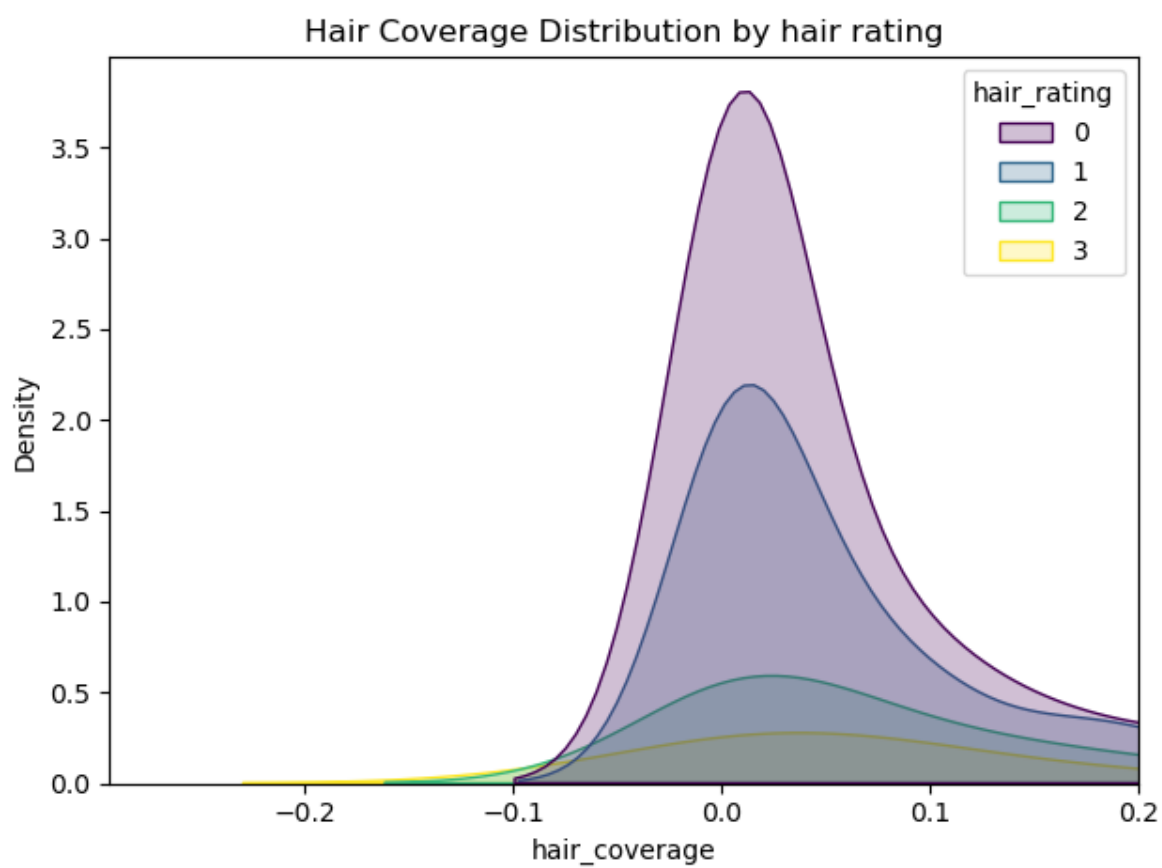


Figure A.17: Distribution of hair coverage across different annotation levels. Higher ratings correspond to increased coverage.

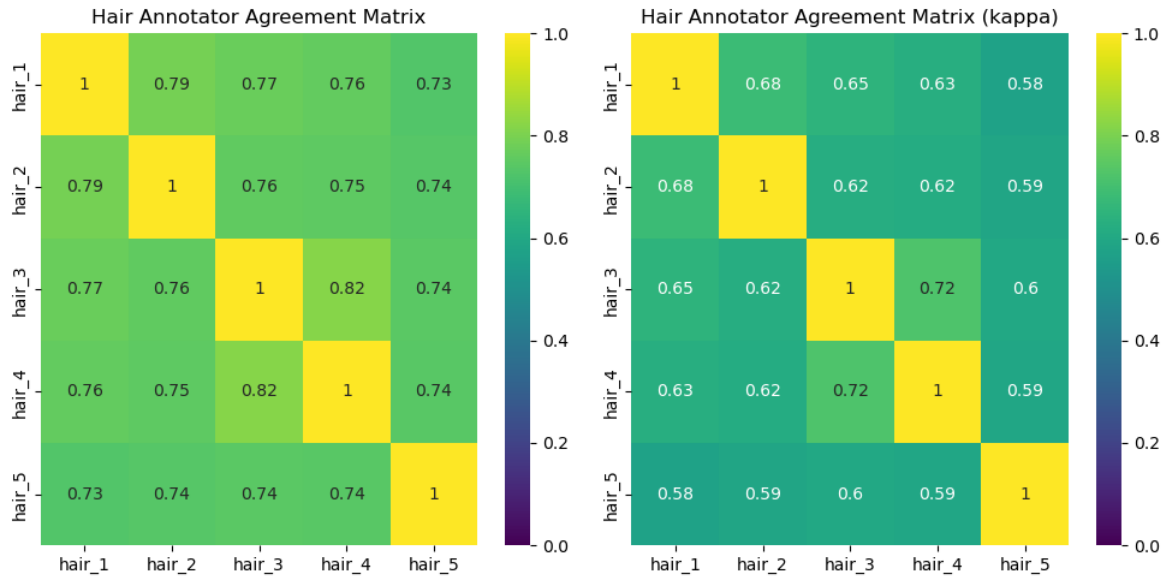


Figure A.18: Hair annotator agreement heatmap.

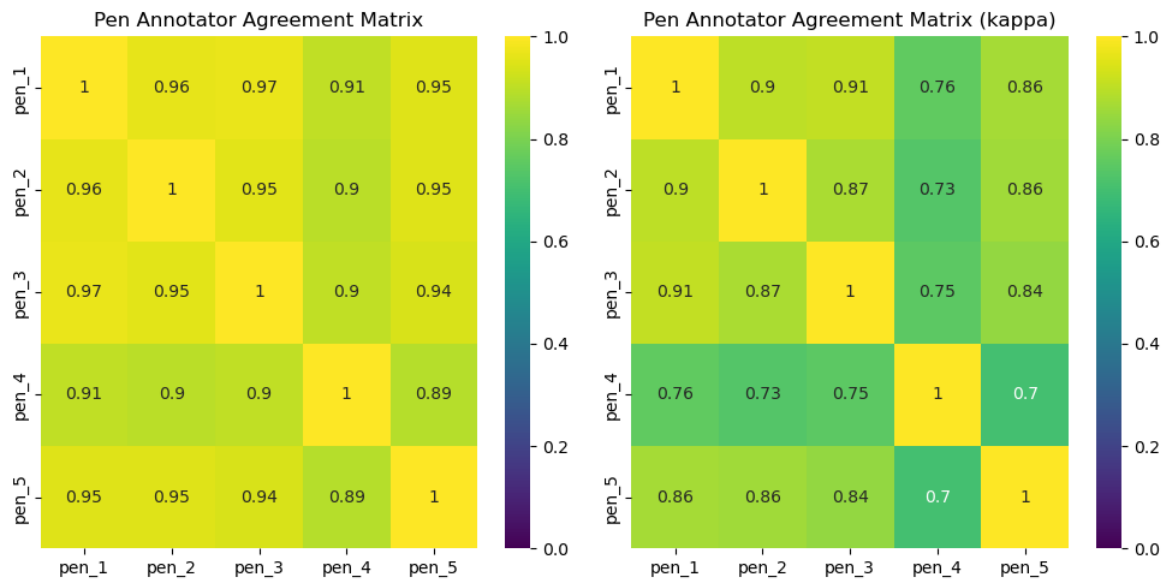


Figure A.19: Pen annotator agreement heatmap.



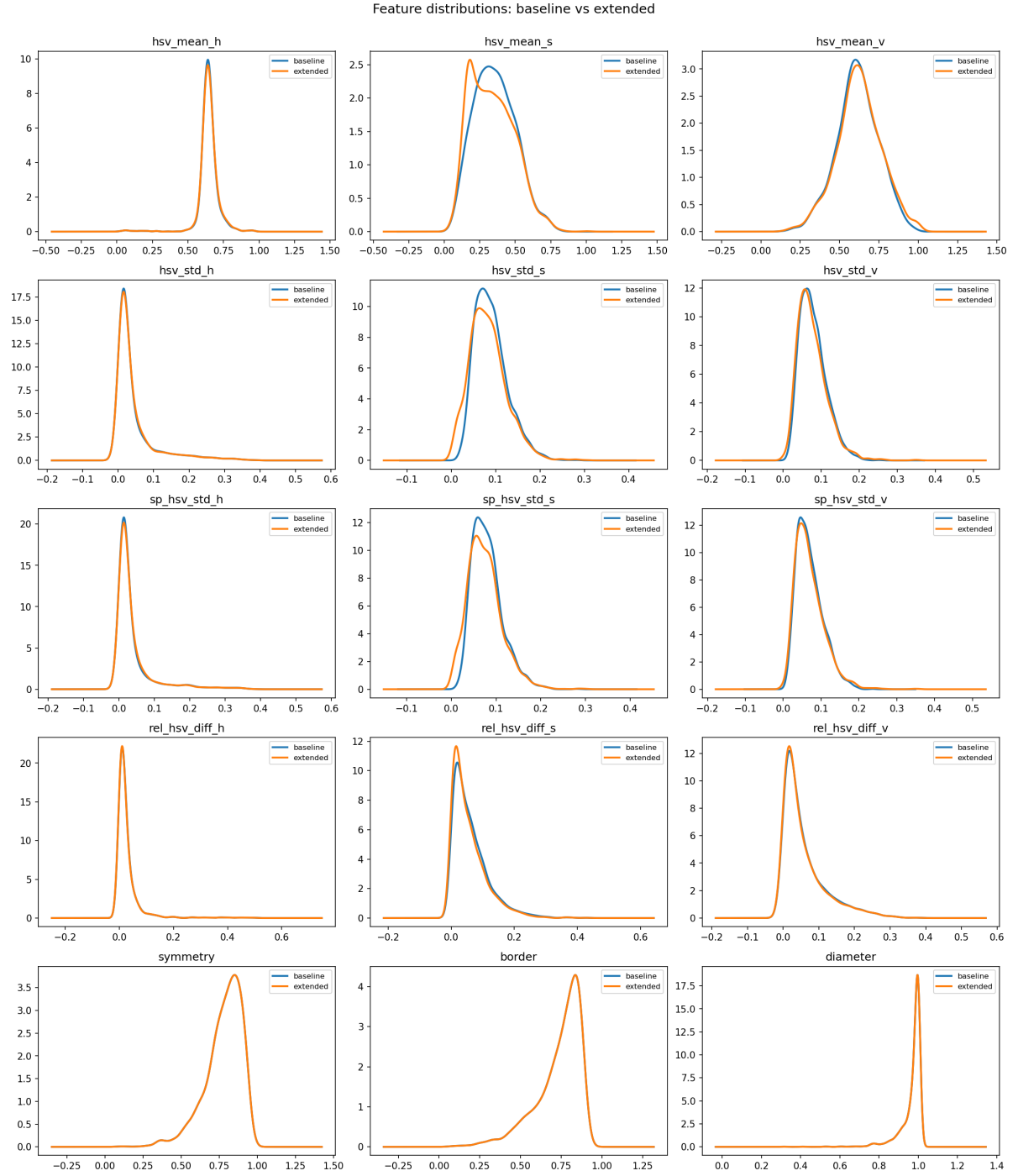


Figure A.21: Comparison of feature distributions for baseline (blue) and extended (orange) models. Color-related features exhibit reduced variability after preprocessing, while shape-based features remain largely unchanged.

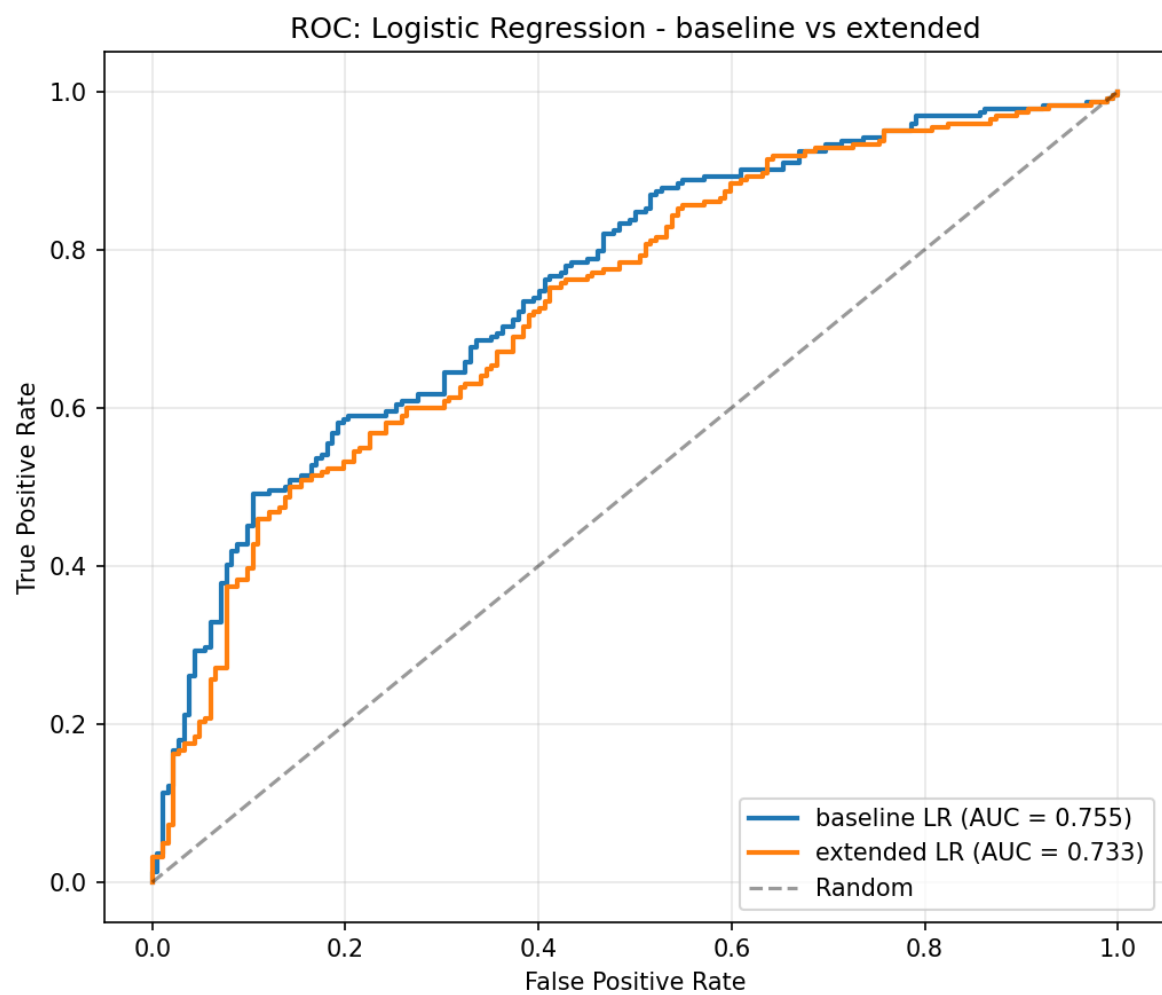


Figure A.22: ROC curves: Logistic Regression trained on baseline vs. extended features.

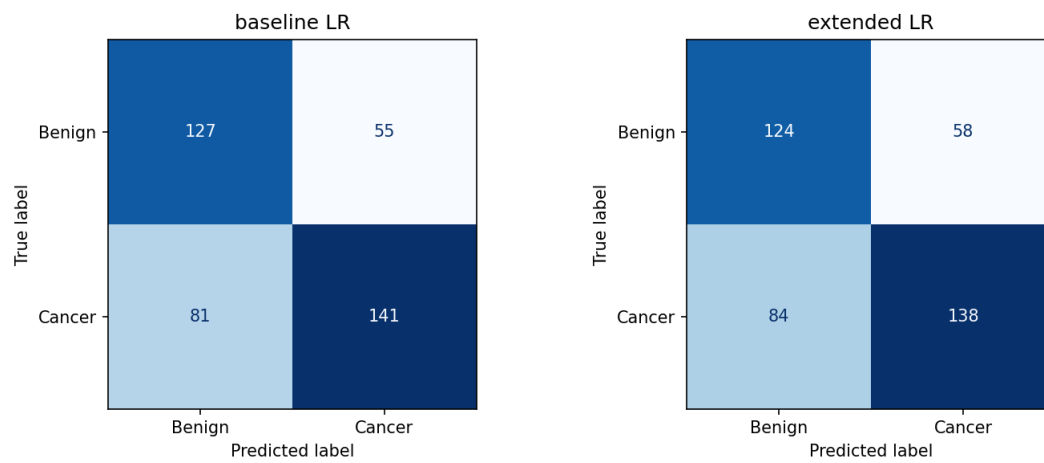


Figure A.23: Confusion matrices for Logistic Regression on the baseline (left) and extended (right) pipelines.

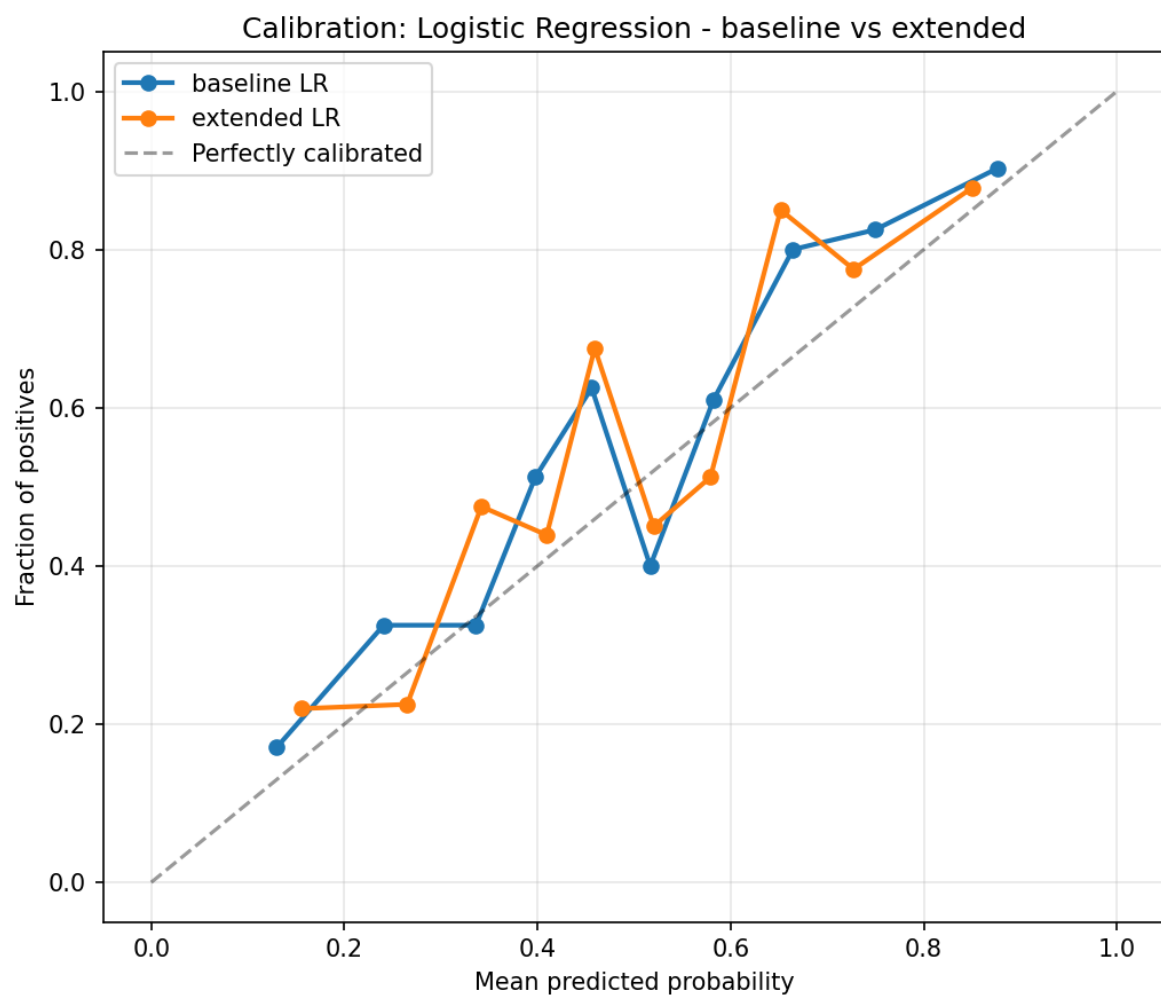


Figure A.24: Calibration plots for Logistic Regression on the two pipelines.

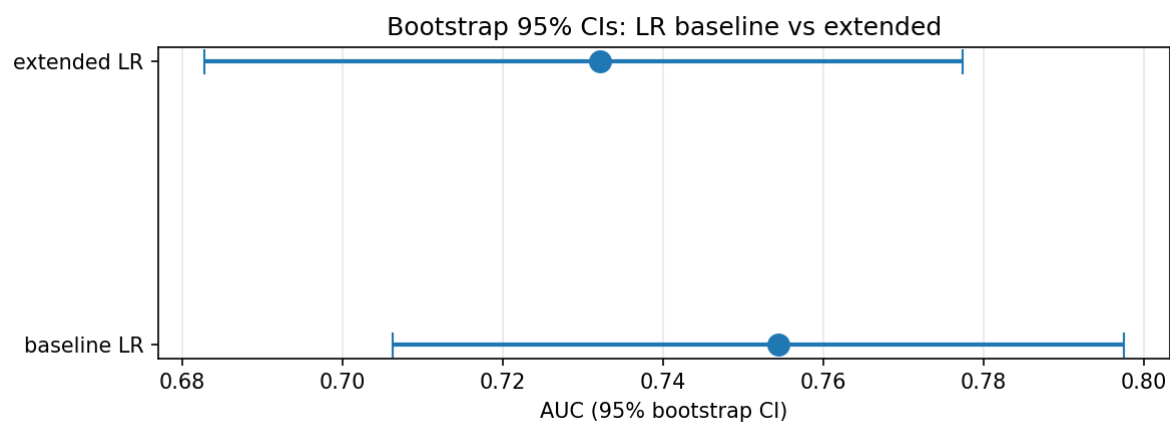


Figure A.25: Bootstrap 95% confidence intervals for test-set AUC: baseline vs. extended LR.

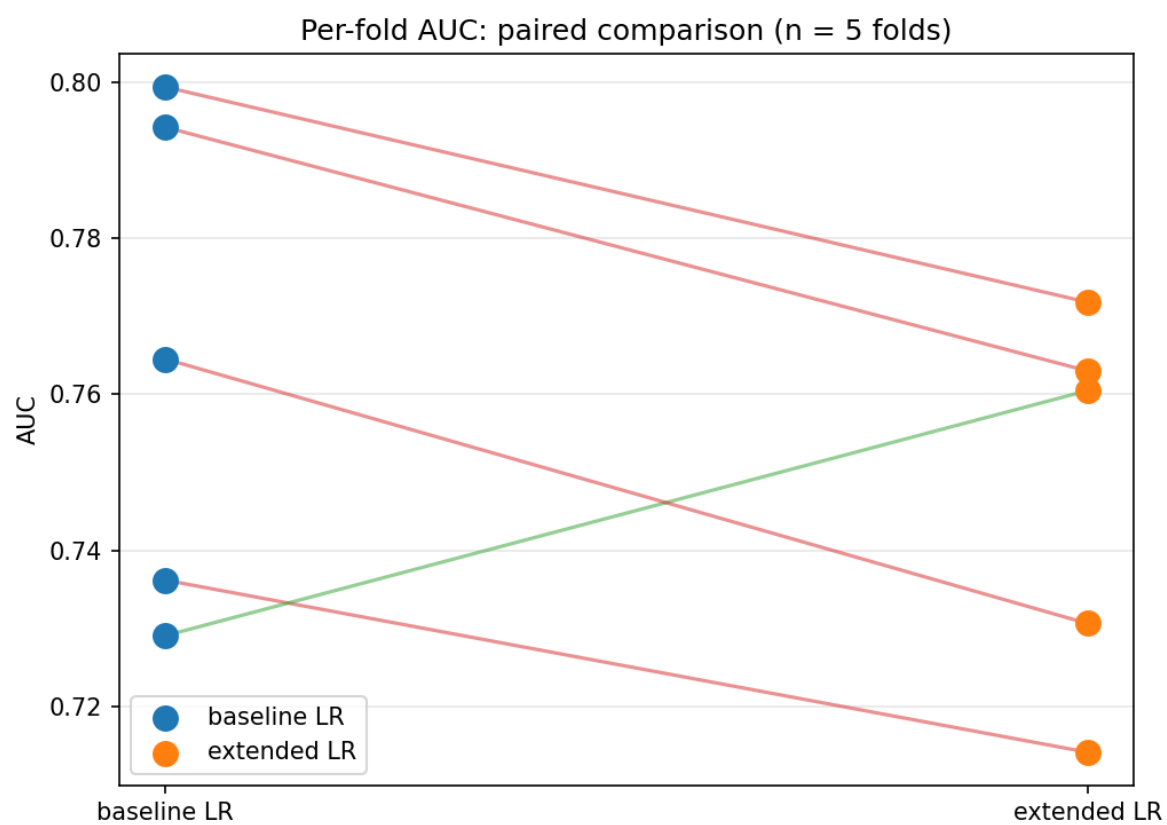


Figure A.26: Per-fold AUC, paired by CV fold. Red lines: extended worse; green: extended better.